



FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Weft Knitting Machines

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CLASSIFICATIONS

A. According to the **frame design** and needle bed arrangement or construction

I. Circular knitting machine

II. Flat knitting machine

B. According to the **number of needle bed** or number of needle set used

I. Single Jersey knitting machine

II. Double Jersey knitting machine

C. According to the **end product** of the weft knitting machine

I. Fabric machine

II. Garment length machine

CLASSIFICATIONS

D. According to the **basic structure** of the weft knitting

I. **Plain** or Single jersey circular knitting machine

II. **Rib** Circular or Flat knitting machine

III. **Interlock** circular knitting machine

IV. **Purl** flat or circular knitting machine

E. According to the **types of needle** used

I. Knitting machine equipped with *Latch needle*

a. Circular knitting machine

❖ One needle bed: Plain, Single-jersey jacquard, Pile and Sliver knit machine

❖ Two needle beds (Dial-cylinder machine): Rib, Interlock, Double-jersey jacquard machine

❖ Double cylinder: Purl knitting machine

CLASSIFICATIONS

b. Flat-bed knitting machine

- ❖ One needle bed: Domestic type
- ❖ Two needle beds: Vee-bed, Flat purl knitting machine

II. Knitting machine equipped with *Spring Bearded needle*

a. Circular knitting machine (Old Machines & No Longer Functioning / Defunct)

- ❖ One needle bed: Sinker wheel, Loop wheel frame.

b. Straight bar frame (All Moderns)

- ❖ One needle bed and Two needle beds: Cotton's Patent or Fully-fashioned machine

TYPES OF MACHINE



S/J Circular m/c



Fleece Circular m/c



Rib Circular m/c



Interlock Circular m/c



Auto-stripe Circular m/c



S/J Jacquard Circular m/c



D/J Jacquard Circular m/c



Flat Bed Knitting m/c

FLAT KNITTING MACHINE



FEATURES OF FLAT KNITTING MACHINE

- ❖ Flat knitting machine has *two stationary needle beds* *Latch needles are used*.
- ❖ *Angular cams of a bi-directional cam* system is used.
- ❖ The cam system is attached to the *underside of a carriage, which with its selected yarn carriers*.
- ❖ The *carriage traverses in a reciprocating manner* across the machine width.
- ❖ There is a *separate cam system for each needle bed*.
- ❖ The *two cam systems are linked together by a bridge*, which passes across from one needle bed to the other.
- ❖ Normally machine *gauge is 3 to 18 needles per inch* and machine *width up to 79 inches*.

FEATURES OF FLAT BED KNITTING MACHINE

- ❖ The machine can be **hand-driven or power-driven**, depending on model and production requirements.
- ❖ **Fabric take-down** is achieved by **rollers or weights** pulling the fabric downward.
- ❖ Flat bed machines are well-suited for **fashion garments, sweaters, collars, ribs, and fully-fashioned** components.
- ❖ The structure of the machine allows **easy patterning** and **individual needle selection** by electronic or mechanical means.
- ❖ **Stitch cams** determine the depth of loop formation, and **tuck or miss cams** can be used for pattern variation.
- ❖ The machine offers **high fabric dimensional stability** and good control over fabric width and shape.

CLASSIFICATIONS OF FLAT KNITTING MACHINE

There are four main types of flat knitting machines:

1. Vee-bed Flat Knitting Machine

It is the most common type of flat knitting machine. It has two needle beds arranged in a V-shape. It is widely used for making: Sweaters, Collars, Cuffs, Other knitted garments. It has a wide range of patterns and shaping possibilities.

2. Flat Purl (Links-Links) Knitting Machine

This machine uses double-headed latch needles. It works in a similar way to the Vee-bed machine. It has: Good pattern-making ability. Better shaping options. Lower production speed compared to some other machines. It is used for different types of knitted fabrics. Both hand-operated and automatic versions are available.

CLASSIFICATIONS OF FLAT KNITTING MACHINE

3. Domestic Single-Bed Flat Knitting Machine

These machines are mainly used for home knitting, not industrial production. They have only one needle bed. Needles are moved by cams attached to a carriage that is moved by hand. Pattern making and shaping are controlled manually. A typical machine has: About 36-inch working width. 5 - 6 needles per inch.

4. Uni-directional Multi-carriage Flat Knitting Machine

This type is made by only one manufacturer. It uses multiple carriages moving in one direction. It is designed for special knitting operations.

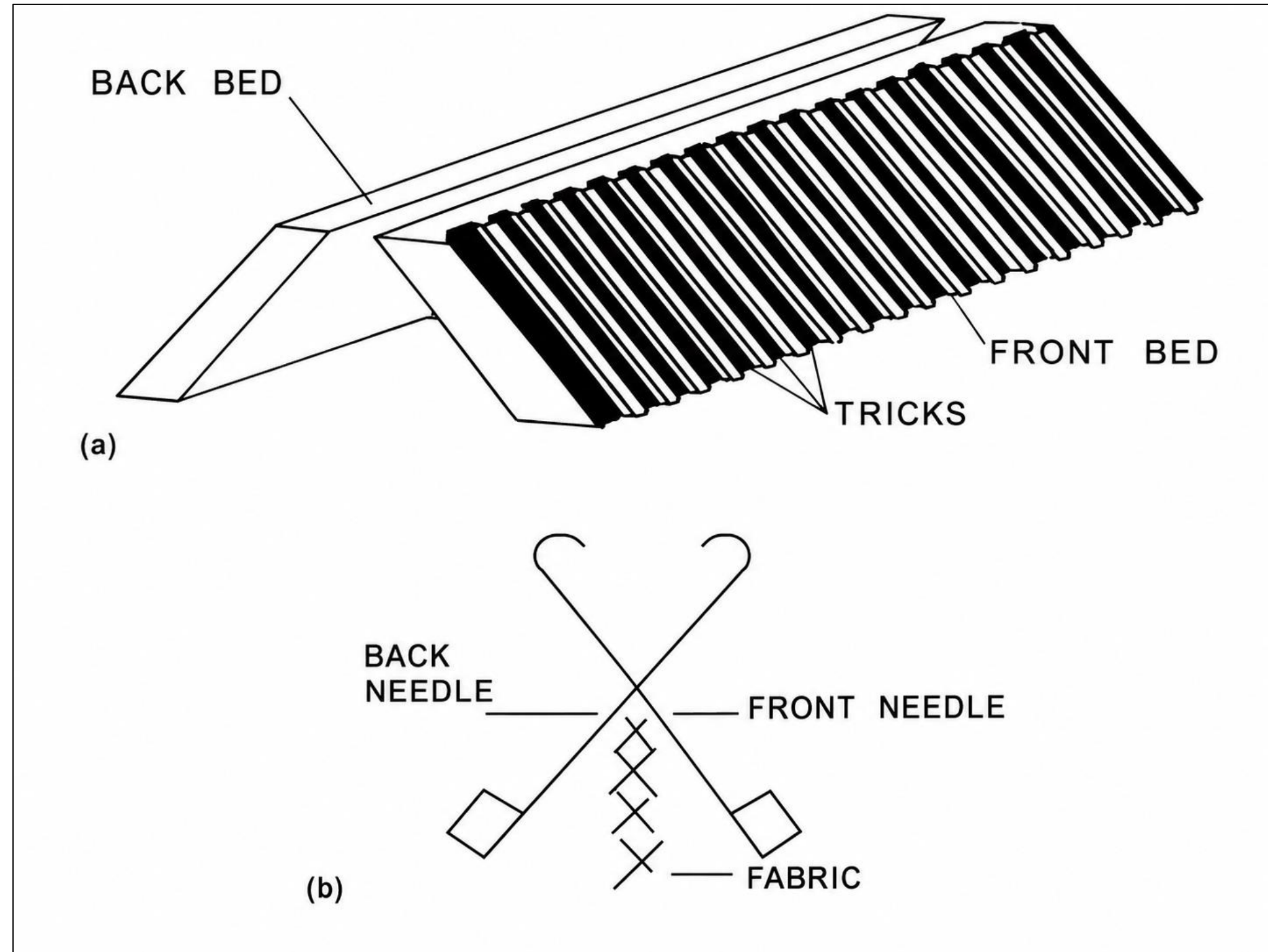
VEE BED FLAT KNITTING MACHINE



FEATURES OF VEE BED FLAT KNITTING MACHINE

- ❖ Vee-bed flat knitting machines are mainly used to produce fully fashioned garments such as: Sweaters, Underwear, Collars.
- ❖ It has two stationary needle beds arranged in an inverted V-shape.
- ❖ The machine is operated by cams in a reciprocating carriage.
- ❖ The front and back needle beds have the same needle capacity.
- ❖ The cam system is movable, so different loop sizes per course can be easily produced.
- ❖ Needles are arranged in rib gating, so rib fabrics can be produced.
- ❖ By changing the cam system, the machine can also produce: Tubular fabrics, Single jersey fabrics.
- ❖ Individual needles can be activated or deactivated by using different mechanisms such as: Springs, Jacks.
- ❖ The machine usually has a coarse gauge range of about E5 to E14.
- ❖ The main disadvantage of this machine is its low productivity compared to other knitting machines.

VEE BED FLAT KNITTING MACHINE



PARTS OF VEE BED FLAT KNITTING MACHINE

1. Frame

- ☐ The frame supports all the needle beds and other necessary parts of the knitting machine.
- ☐ It provides a strong base for the knitting process.

2. Needle Bed

- ☐ The machine has stationary front and back needle beds.
- ☐ The needle beds hold the needles in the correct position.
- ☐ Spring securing bars and needle securing bars are attached to the needle bed to keep needles fixed.

3. Carriage

- ☐ The carriage consists of two metal plates connected by a bridge.
- ☐ The plates work separately and together on the front and back needle beds.
- ☐ It controls the needles by: Selecting needles, Moving needles up and down, Forming stitches.
- ☐ Driving yarn guides that feed yarn to the needles.

PARTS OF VEE BED FLAT KNITTING MACHINE

4. Brushes

- ☐ Brushes are mounted on the carriage.
- ☐ They help in: Smooth opening of needle latches, Proper feeding of yarn to the needles.

5. Cam Locks

- ☐ The movable cam system provides the required movement information to individual needles.
- ☐ It controls needle selection and stitch formation.

6. Latch Needles

- ☐ Two types of latch needles are used: High butt needles, Low butt needles.
- ☐ They help to intermesh loops and pull new loops through old loops.

PARTS OF VEE BED FLAT KNITTING MACHINE

7. Cover Plate

- ☐ Thin metal blade located in a slot across the top of the needle bed tricks.
- ☐ It prevents the stems of the needles from pivoting upward out of the needle trick whilst allowing the needles to slide freely in their trick.
- ☐ The plate can be withdrawn sideways out of the needle bed to allow damaged needles to be replaced.

8. Security Spring:

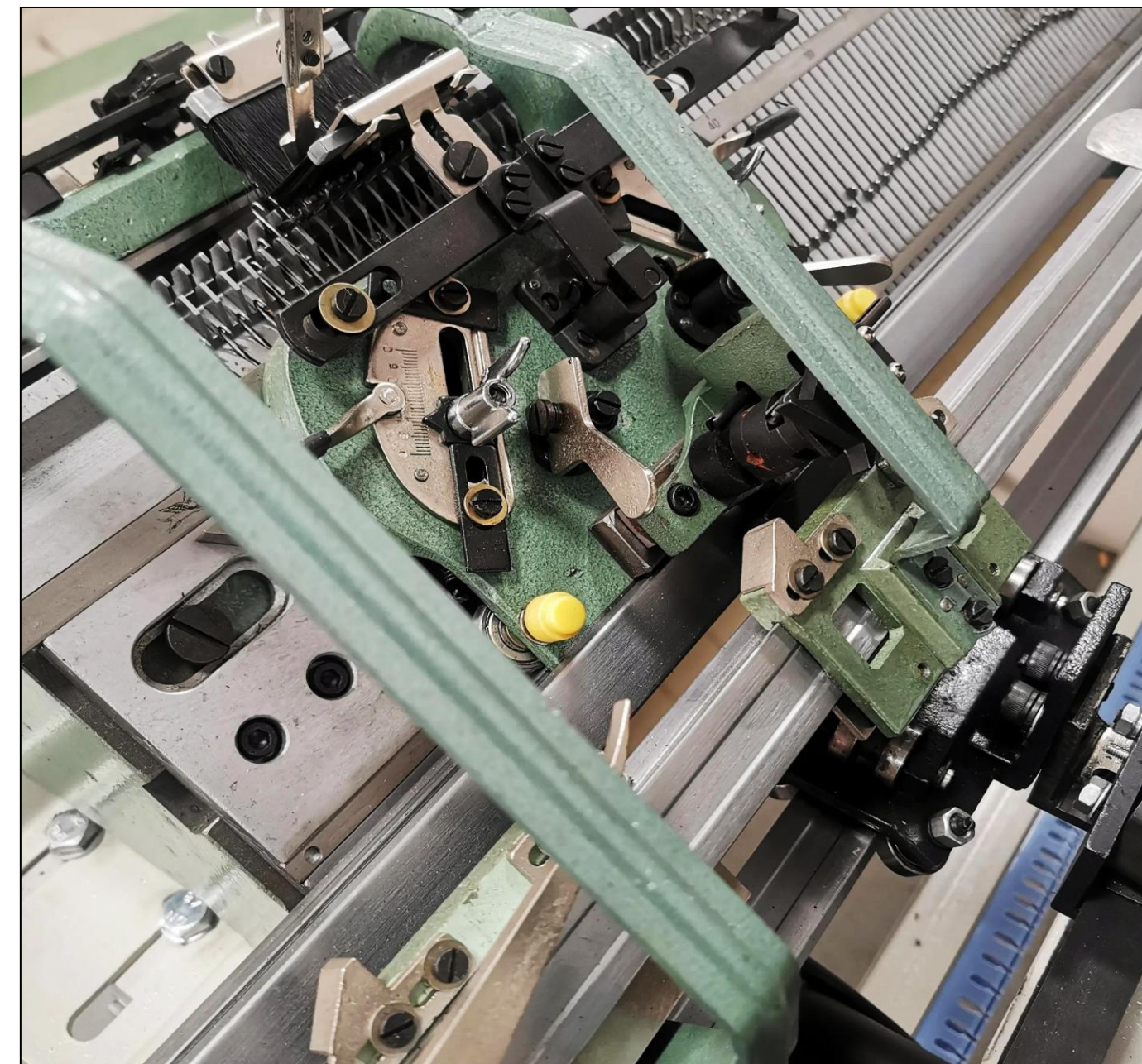
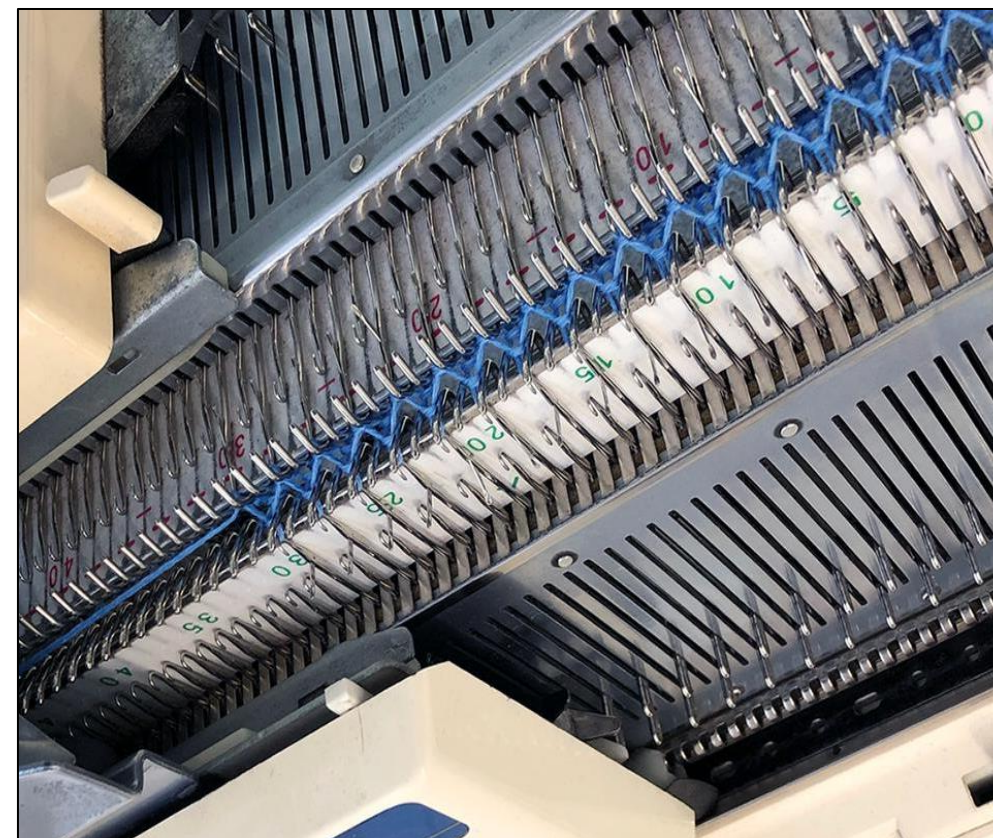
- ☐ Supporting the tail of each needle is a security spring that fits over the lower edge of the needle bed.
- ☐ When spring is pushed fully into the position, it locates into a groove on the under-surface of the needle bed.
- ☐ The butt of the needle bed that it supports is then aligned with the knitting cam track on the under surface of the traversing cam carriage.
- ☐ When a needle is not required to be in action its security spring isn't located in the groove, so that needle is nearer to the lower edge of the needle bed and its butt misses the traversing cam carriage.

PARTS OF VEE BED FLAT KNITTING MACHINE

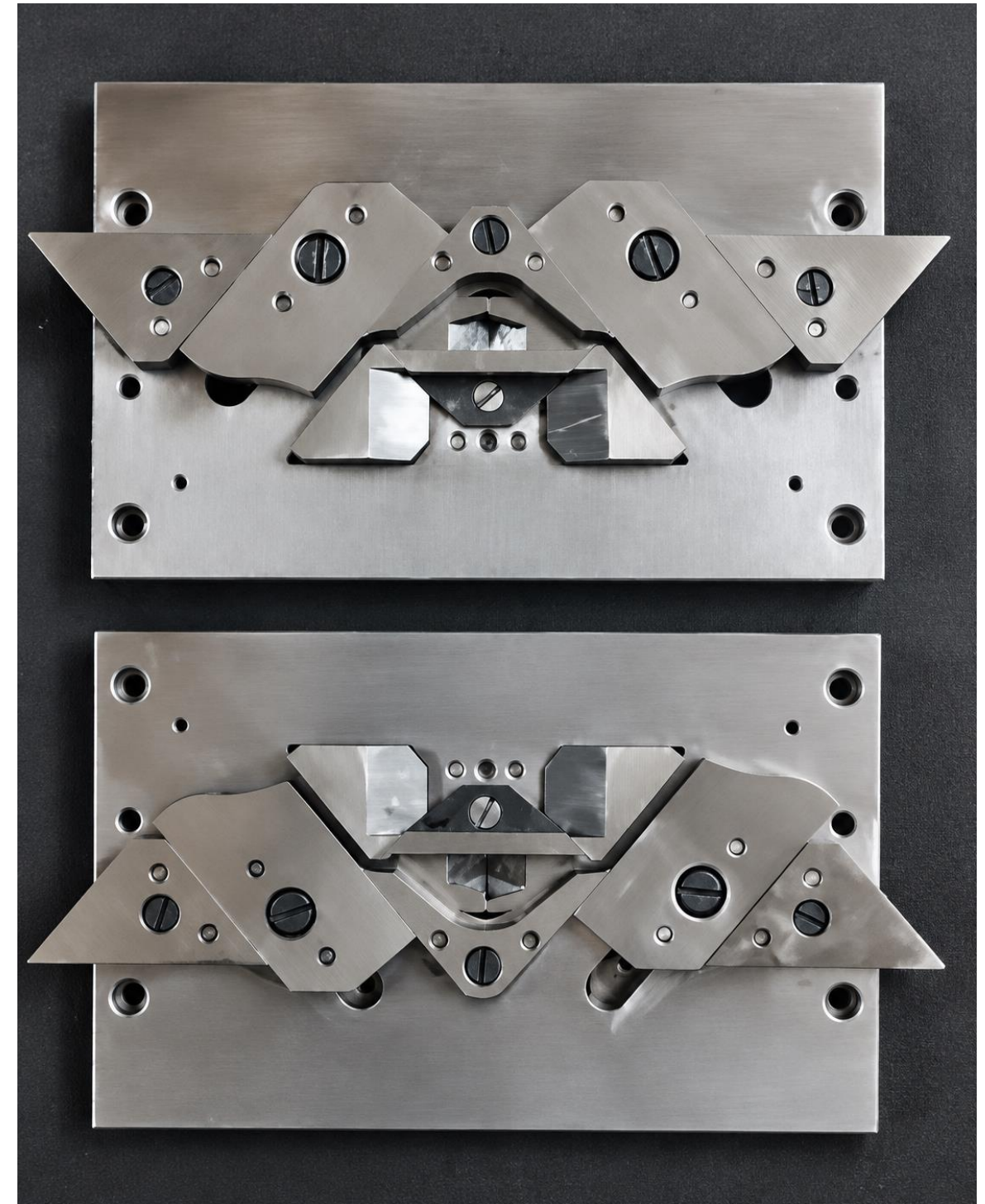
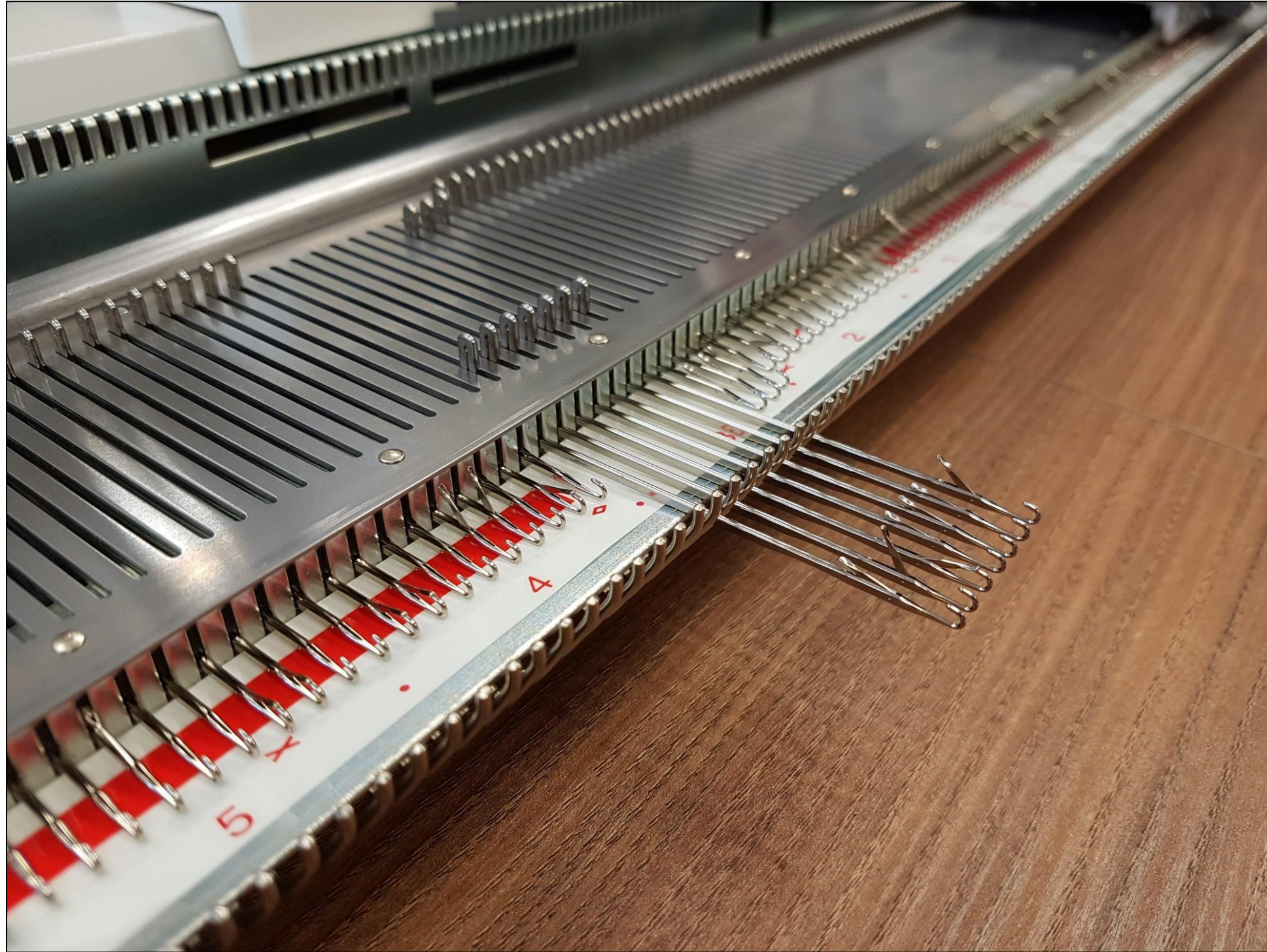
9. Fabric Take-Down Comb

- ❑ A special system is used to remove the knitted fabric from the machine.
- ❑ After the first carriage stroke, a fabric comb is hand inserted into it, upwards from under the needle bed, so that the eyelet holes of the comb protrude above the course.
- ❑ Comb wire is then inserted through the eyelets, over the set up course, so that the comb is suspended from the course and a takedown weight is attached to it..
- ❑ The weight creates tension to pull the fabric downward.
- ❑ Machine gauge = Needle gauge.

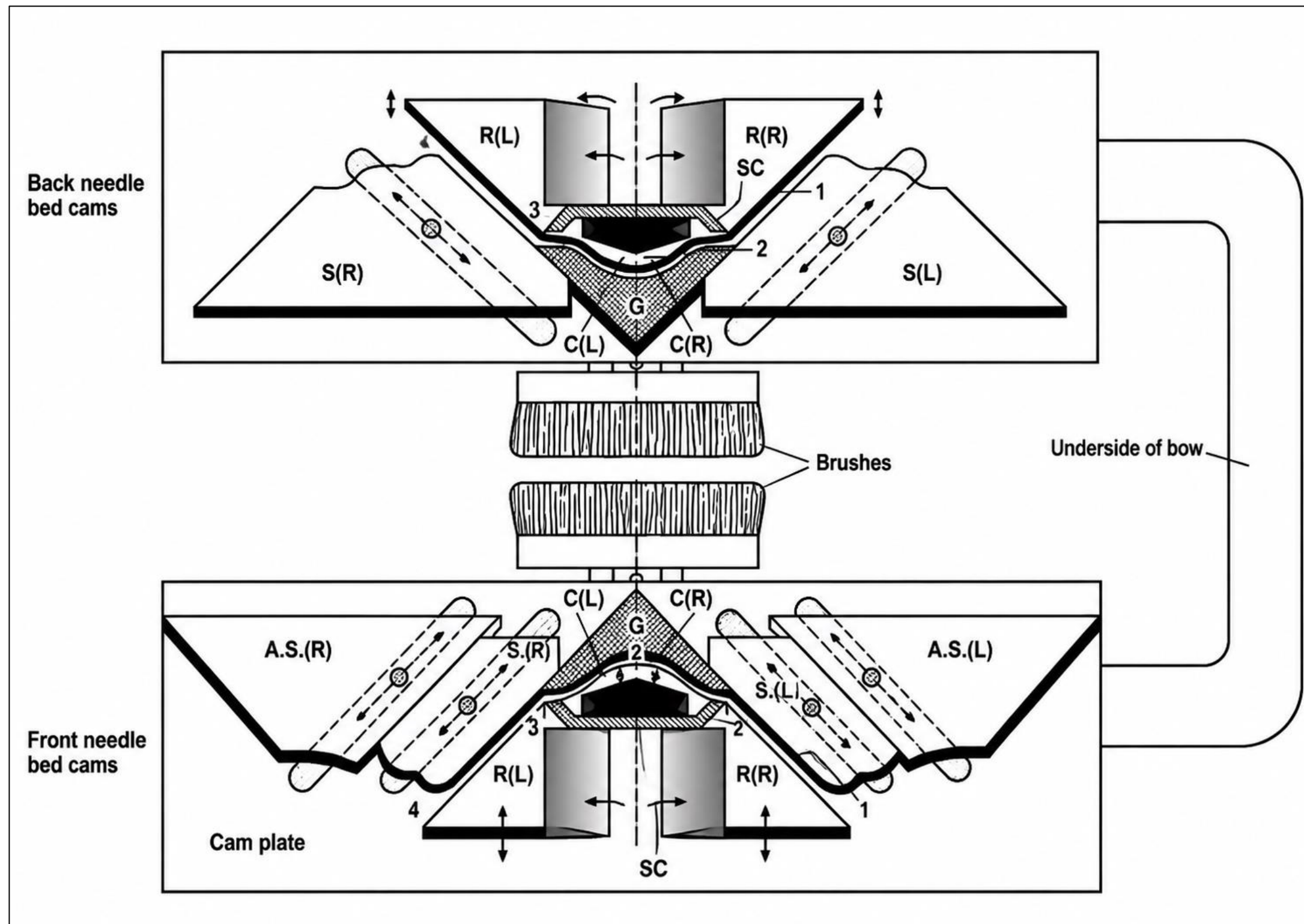
PARTS OF VEE BED FLAT KNITTING MACHINE



PARTS OF VEE BED FLAT KNITTING MACHINE



PARTS OF VEE BED FLAT KNITTING MACHINE



- ❖ R(R): Raising Cam (Right)
- ❖ S(R): Stitching Cam (Right)
- ❖ C(R): Cardigan Cam (Right)
- ❖ R(L): Raising Cam (Left)
- ❖ S(L): Stitching Cam (Left)
- ❖ C(L): Cardigan Cam (Left)
- ❖ G: Guard Cam

PARTS OF VEE BED FLAT KNITTING MACHINE

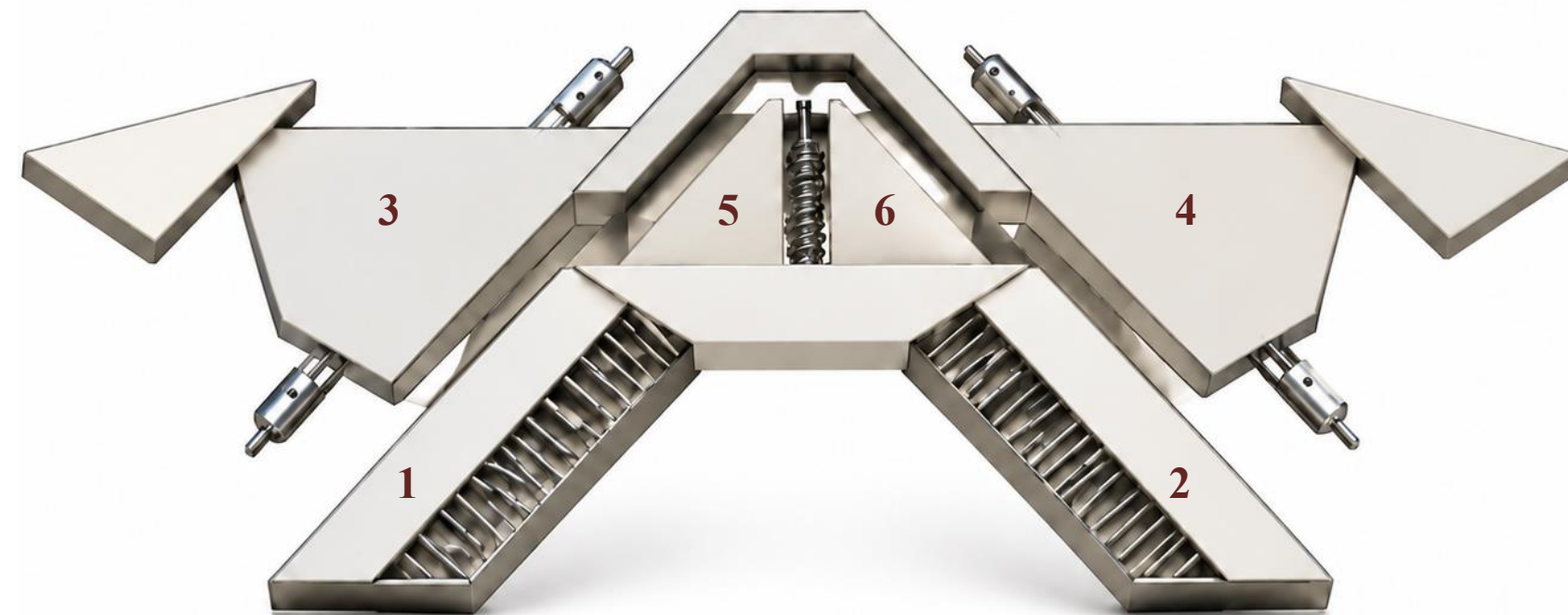
1,1',2,2' – Raising Cam

3,3',4,4' – Stitch Cam

5,5',6,6' – Cardigan/Tuck Cam



BACK BED



FRONT BED

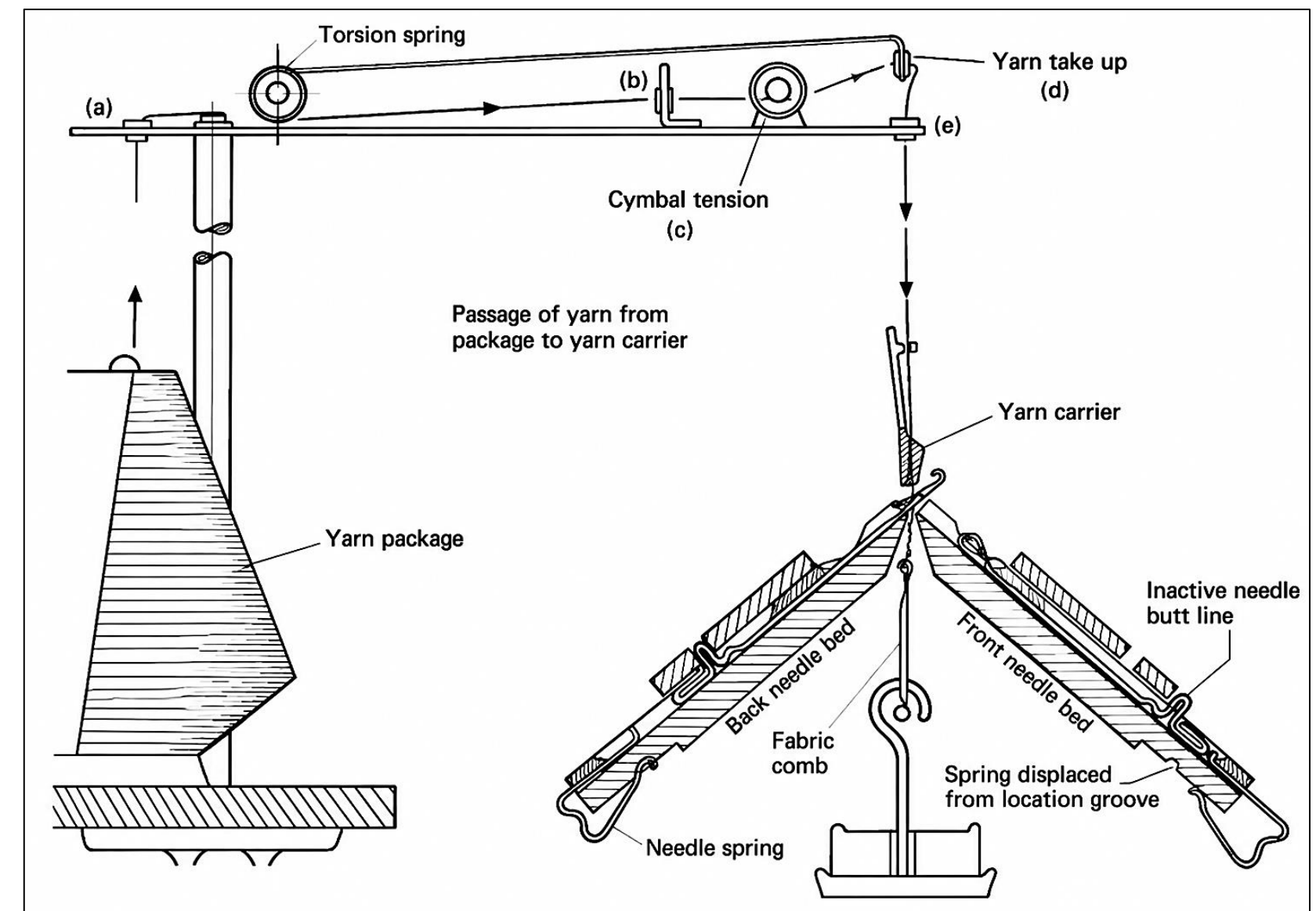
YARN PATH OF VEE BED FLAT KNITTING MACHINE

In the rear side of the m/c, there is a spool rack for storing the yarn spools. The yarn unwound from the spool is picked up by the feeding system:

Tensioner: Which adjusts the feeding tension. The tensioner is made up by a spring load flexible arm, that **lowers when the yarn tension increases** to feed a bigger quantity of yarn, and **lifts up when the tension decreases**.

Thread Guide: Which provides the needle with yarn at the proper time.

Carriage System: Carries the yarn feeders, yarn, cam boxes, brush etc.

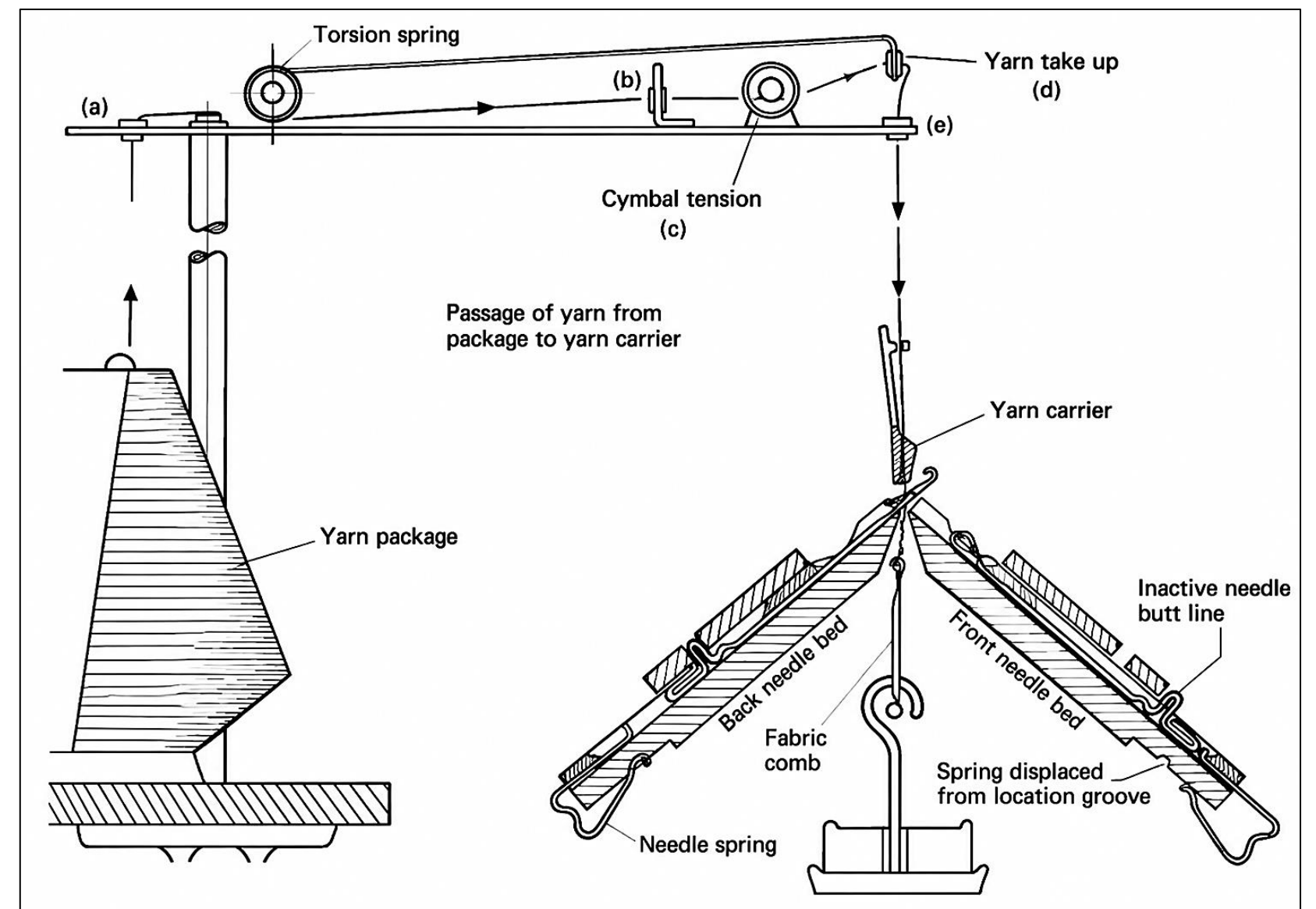


YARN PATH OF VEE BED FLAT KNITTING MACHINE

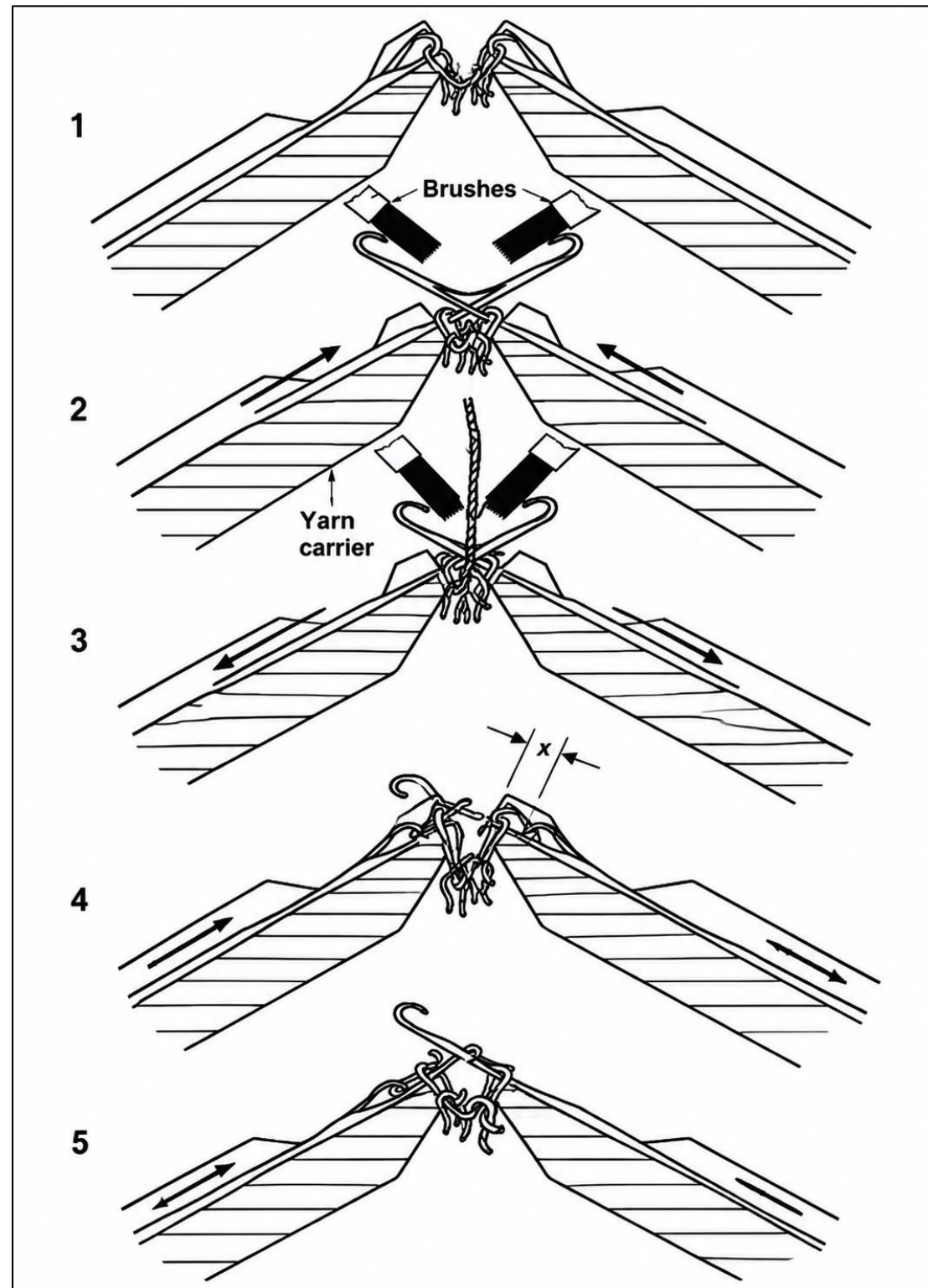
Latch Opening Brushes: Are attached to the cam-plates of both needle beds to ensure that the needle latches are fully opened.

Trick Walls: Are replaced at the needle bed verges by fixed, thinner, polished and specially shaped knock-over bit edges. Latch needles are used. There is a security spring that fits over the lower edge of the needle bed.

Fabric Take-down Comb: It's a hand-inserted upwards from under the needle beds, so that the eyelet holes of the comb protrude above the course. The comb wire is then inserted through the eyelets, and a takedown weight is attached to it.



VEE BED FLAT KNITTING MACHINE **KNITTING ACTION**



- ❖ Position 1: Rest-in
- ❖ Position 2: Clearing
- ❖ Position 3: Yarn Feeding
- ❖ Position 4: Knocking Over
- ❖ Position 5: Tucking-in

VEE BED FLAT KNITTING MACHINE **KNITTING ACTION**

Position 1: Rest-in

The tops of the heads of the needles are level with the edge of the knock over bits. The butts of the needle assume a straight line until contacting the raising cam leading stitch cams are lifted to an inactive position. The lifting action is an alternating action that always lowers the trailing stitch cams and raises the leading stitch cams in each system as the traverse commences. This action prevents needles from being unnecessarily lowered and a strain being placed on the old loops prior to the start up of the knitting action.

Position 2: Clearing

The needle butts are lifted as they contact the leading edge of cams that raise the needle to “tucking in the hook” height with the under surface of cams acting as guard cams. The needles are lifted to full clearing height as their butts pass over the top of the tuck cams.

VEE BED FLAT KNITTING MACHINE **KNITTING ACTION**

Position 3: Yarn Feeding

The yarn is fed as the **needles descend** under the control of guard cam. Each needle draws the required loop length as it descends the stitch cam.

Position 4: Knocking Over

To produce synchronize knocking over of both needle bed simultaneously; the stitch cam front system is set lower the auxiliary stitch cam so that the later is rendered ineffectively. If, however delayed timing of knock over is employed, knock over in front bed will occur after knock over in the back bed. In this case, stitch cam is not set as low as fig-5 so that the depth setting of the later cam produce the knock over action. Delayed timing is only normally used on gauges finer than 8NPI (Needles Per Inch) and cannot be used for broad ribs.

VEE BED FLAT KNITTING MACHINE LOOP FORMATION

Knit stitch, tuck stitch, and miss stitch can be formed by activating or deactivating the raising and cardigan cams.

Knit stitch: When the needle carries out a complete stroke, reaching the maximum height on the looping plane.

Tuck stitch: When the needle reaches the tucking plane and receives a new yarn while still holding its former loop, thus forming two loops in one needle hook.

Miss or float stitch: When the needle is not knitting and remains out of the knock-over plane.



SINGLE JERSEY TUBULAR FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Tubular knitting is also known as circular knitting. It produces a tube-shaped fabric. It can be made using:

- ✓ Circular knitting machines with one needle bed.
- ✓ Flat knitting machines with two needle beds (front and back beds). In flat machines, the fabric is knitted as a flat tube and takes a circular shape after knitting.

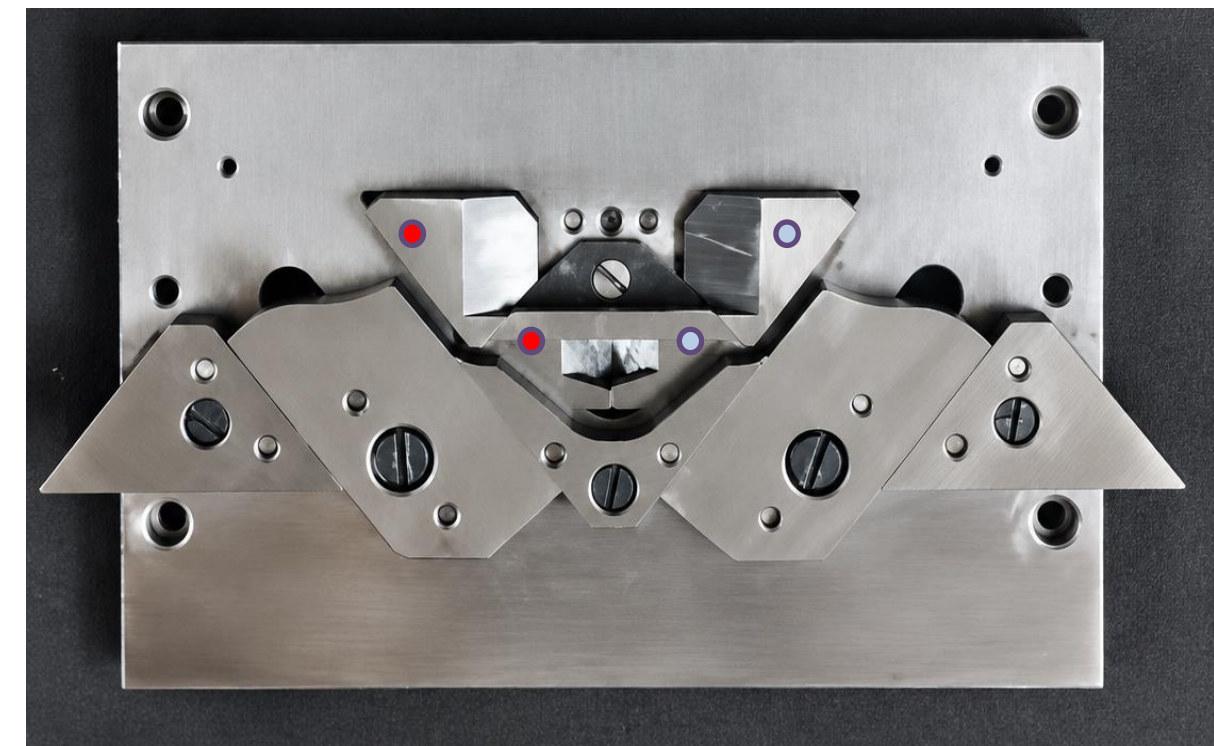
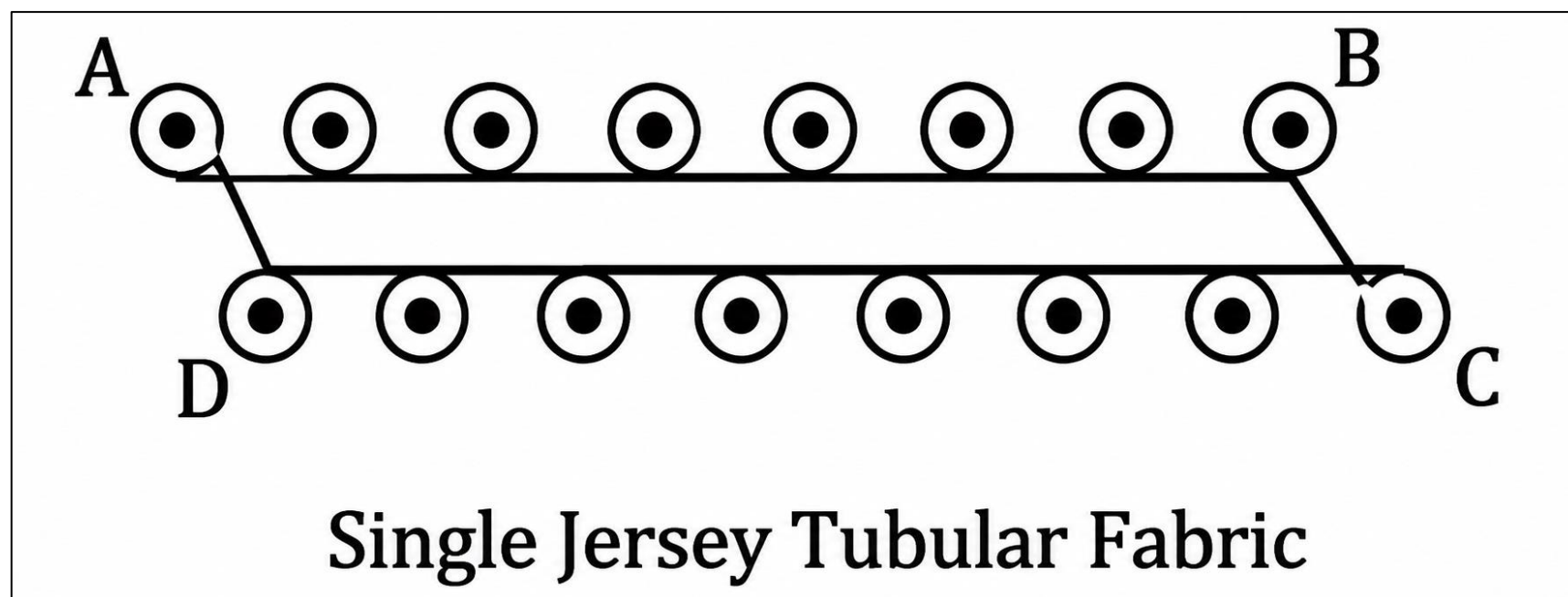
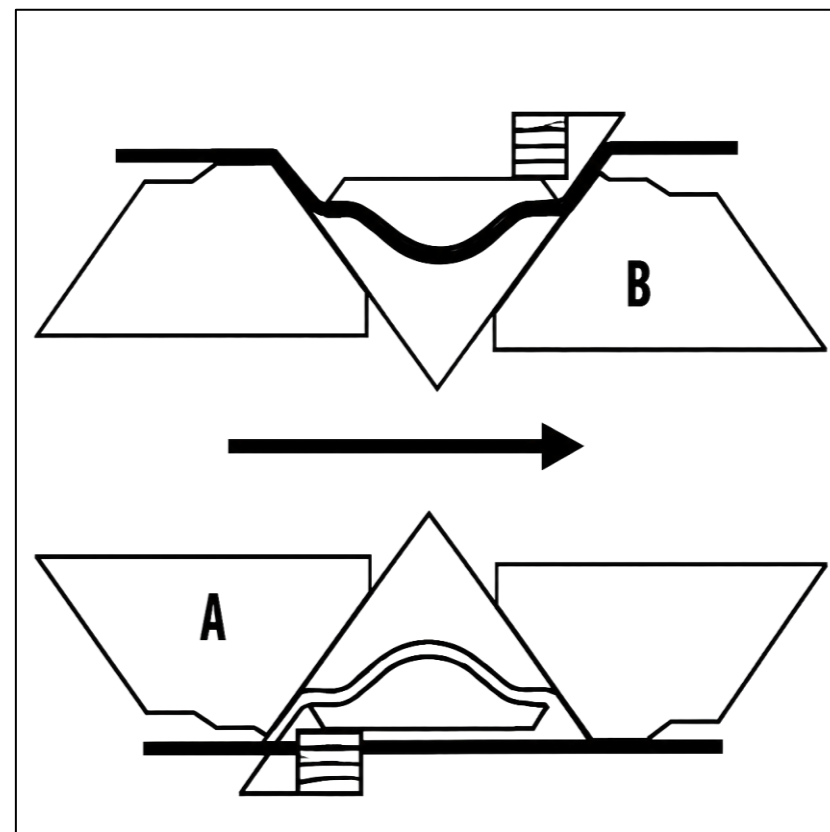
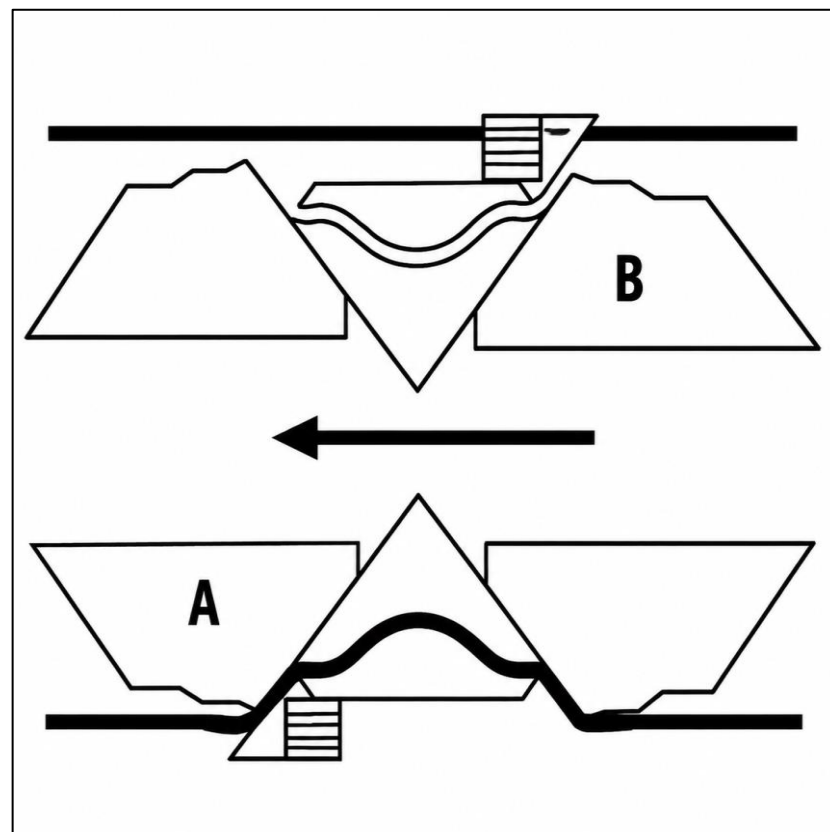
Open Tubular Knitting:

- If the *front and back fabrics are not joined at the edges*, an open tubular fabric is produced.

For one-side connection:

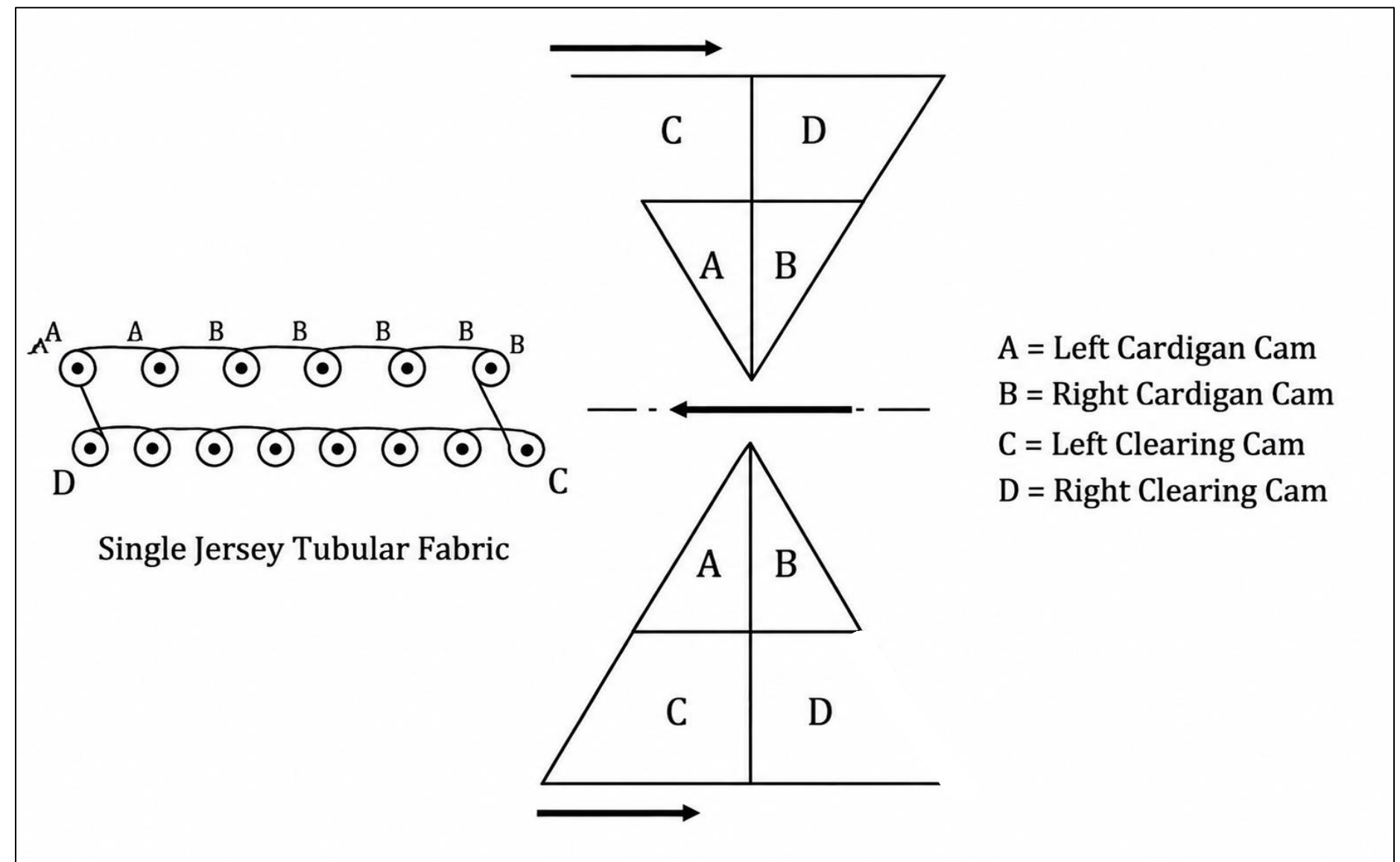
- Knit two rows on the back needle bed.
- Then knit two rows on the front needle bed.
- The joining occurs only on one side.

SINGLE JERSEY TUBULAR FABRIC PRODUCTION IN FLAT KNITTING MACHINE



SINGLE JERSEY TUBULAR FABRIC PRODUCTION IN FLAT KNITTING MACHINE

- ❖ Two raising cams, one at the front, one at the back (C & D) are out of action.
- ❖ When the carriage travels from left to right, the back needles only will form a stitch.
- ❖ On the other hand, when the carriage goes from right to left, only the front needles form a stitch.
- ❖ The wales of the selvedges AD and BC will be joined.



SINGLE JERSEY FLAT FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Single bed fabric is also known as jersey fabric. It is produced by *using only one needle bed (usually the back needle bed)*.

Characteristics:

1. Lightweight fabric.
2. Has low elasticity compared with other knitted structures.
3. Stitches are formed uniformly.
4. Fabric quality depends on: Proper needle bed condition, smooth and polished needle parts, and accurate setting of lowering cams.
5. Incorrect cam setting can cause uneven stitch length, resulting in visible stripes on the fabric.

SINGLE JERSEY FLAT FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Usually, single bed fabric is started after *making a rib fabric welt* using two needle beds.

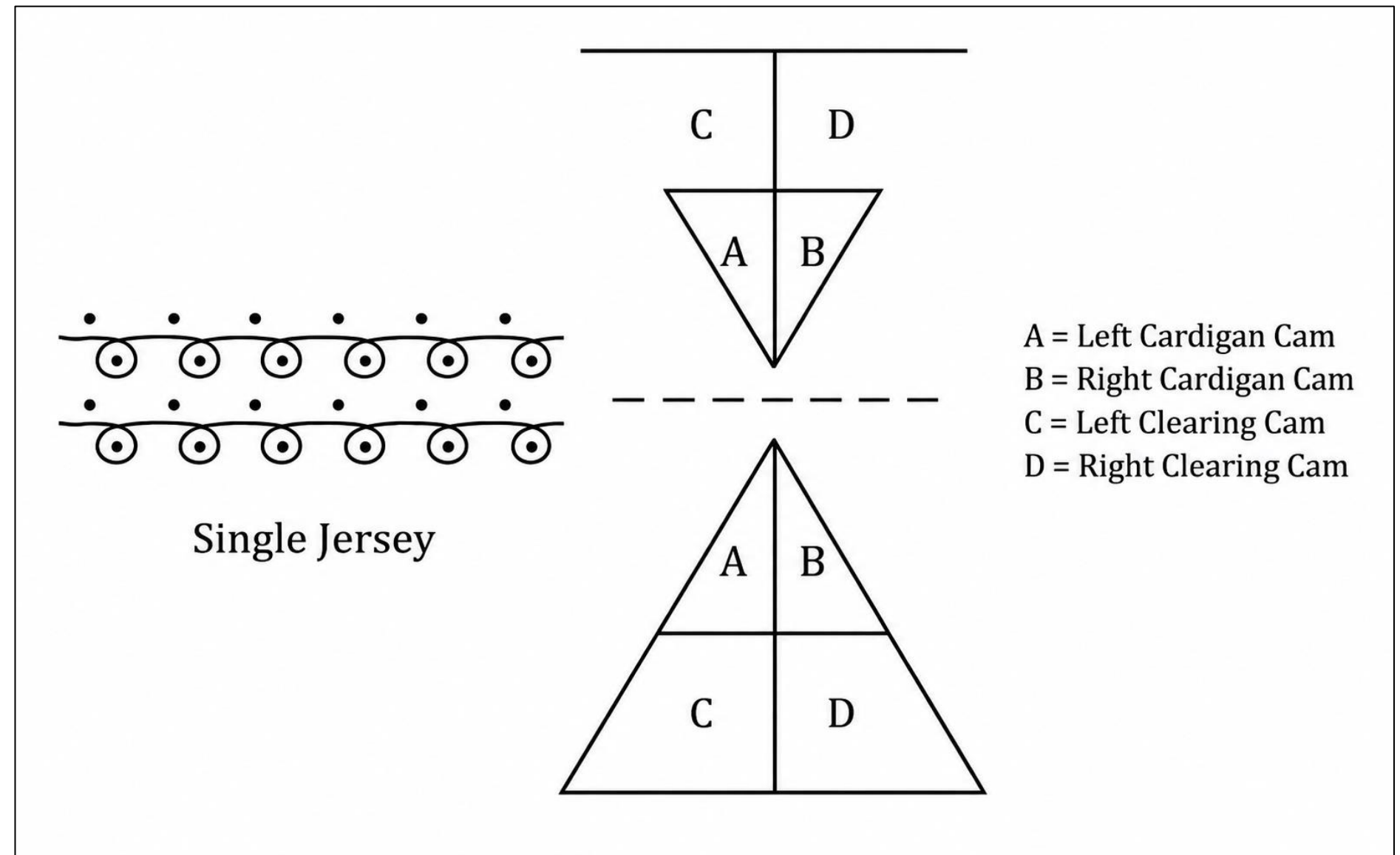
- ❖ To change *from rib fabric to single bed* fabric, *all stitches from one needle bed are transferred to the other needle bed* using a hook (usually front to back bed).
- ❖ After transferring the stitches, knitting continues using only one needle bed.

However, single bed fabric can also be started without a rib welt.

- ❖ In this case, a set-up comb is placed between the two needle beds.
- ❖ During the first knitting row, the *needles pass through the comb openings and catch the yarn*.
- ❖ *The yarn becomes held by the comb*, allowing single bed knitting to start.

SINGLE JERSEY FLAT FABRIC PRODUCTION IN FLAT KNITTING MACHINE

- ❖ Both raising cams, at the back (C & D) are out of action.
- ❖ All cams at the front bed are in action.
- ❖ When the carriage travels from left to right, the front needles only will form knit stitch.



SINGLE JERSEY FLAT FABRIC PRODUCTION IN FLAT KNITTING MACHINE [LOOP TRANSFER]

GROZ-BECKERT

Loop formation and stitch transfer: flat knitting

Spring-loaded latch needle

RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Unlike circular fabric or single bed fabric, rib fabric is produced using both needle beds simultaneously. The yarn moves alternately between the front and back needle beds, forming the rib structure.

- 1×1 rib fabric has the same appearance on both sides.
- It is highly elastic, especially in the width direction.
- Elasticity depends on: Fabric structure, Stitch length, Type of yarn
- In 1×1 rib knitting, *all needles of both needle beds are active.*
- Therefore, *all raising cams work, and lowering cams must be set at the same position for uniform stitch formation.*

RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Rib fabric is a knitted fabric produced using two needle beds (front and back needle beds). In rib knitting, some needles are kept active and some needles are kept inactive to create different rib structures.

Rib fabrics are mainly used for: Pullover bottom borders, Jacket borders, Sleeve cuffs because they are elastic and close-fitting.

1. 2/3 Rib Fabric (2×2 Rib)

- In 2/3 rib, 2 needles out of 3 are active. 1 needle remains inactive.
- The inactive needle is placed between the active needles of the opposite needle bed.
- Good width-wise elasticity.

RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

2. 1/2 Rib Fabric (2×1) Rib

- In 1/2 rib, one needle is active and one needle is inactive alternately.
- It has higher elasticity than 2/3 rib.

3. 2/4 Rib Fabric (3×2) Rib

- In 2/4 rib, two needles are active followed by two inactive needles.
- It provides even higher elasticity than 2/3 rib.

RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

Inactive Needle

An inactive needle does not participate in knitting. The needle is pushed down to the bottom of the needle bed. Its butt remains below the raising cam, so it does not work.

Elasticity of Rib Fabric

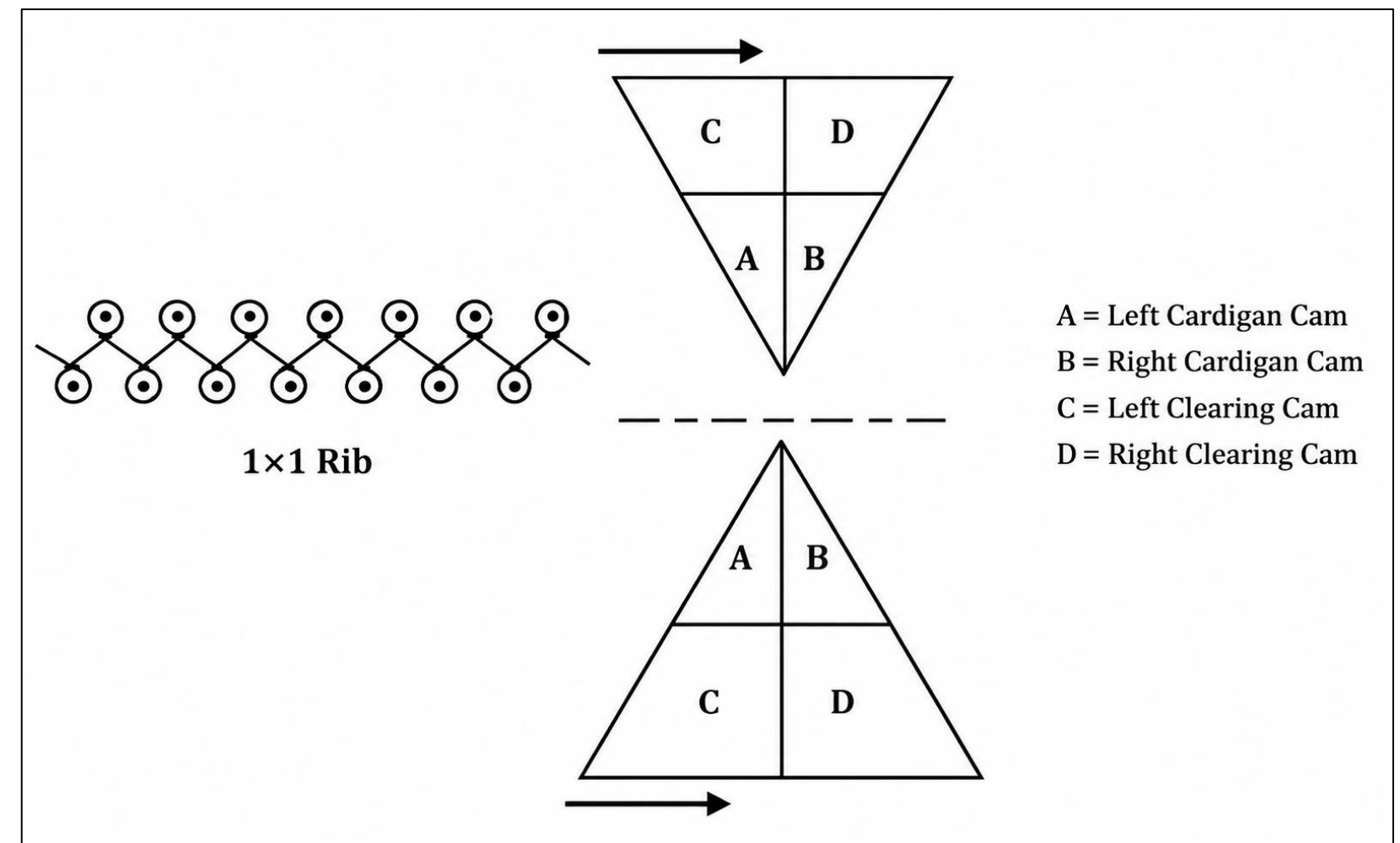
Rib fabrics are highly elastic in the width direction.

The elasticity occurs because:

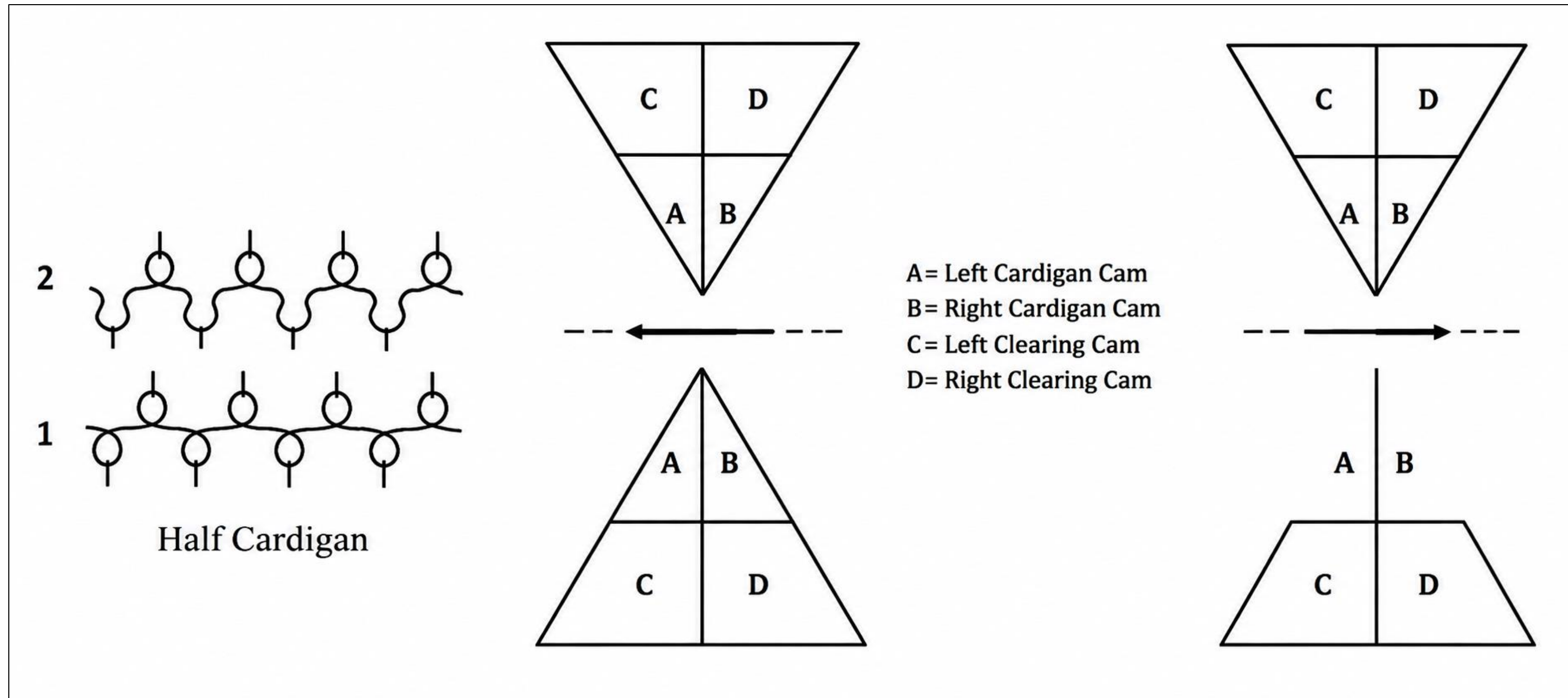
- *Inactive needles create extra spaces* in the structure.
- These *spaces allow the ribs to close and expand easily*.
- *Single stitch rib structures generally show higher elasticity*.

1x1 RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

- ❖ *All raising cams & all lowering cams will be in action.*
- ❖ When the carriage will travel from left to right, both front & back bed needles will form knit stitch.
- ❖ When the carriage travel from right to left, both front & back bed needles will form knit stitch.



HALF CARDIGAN RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE



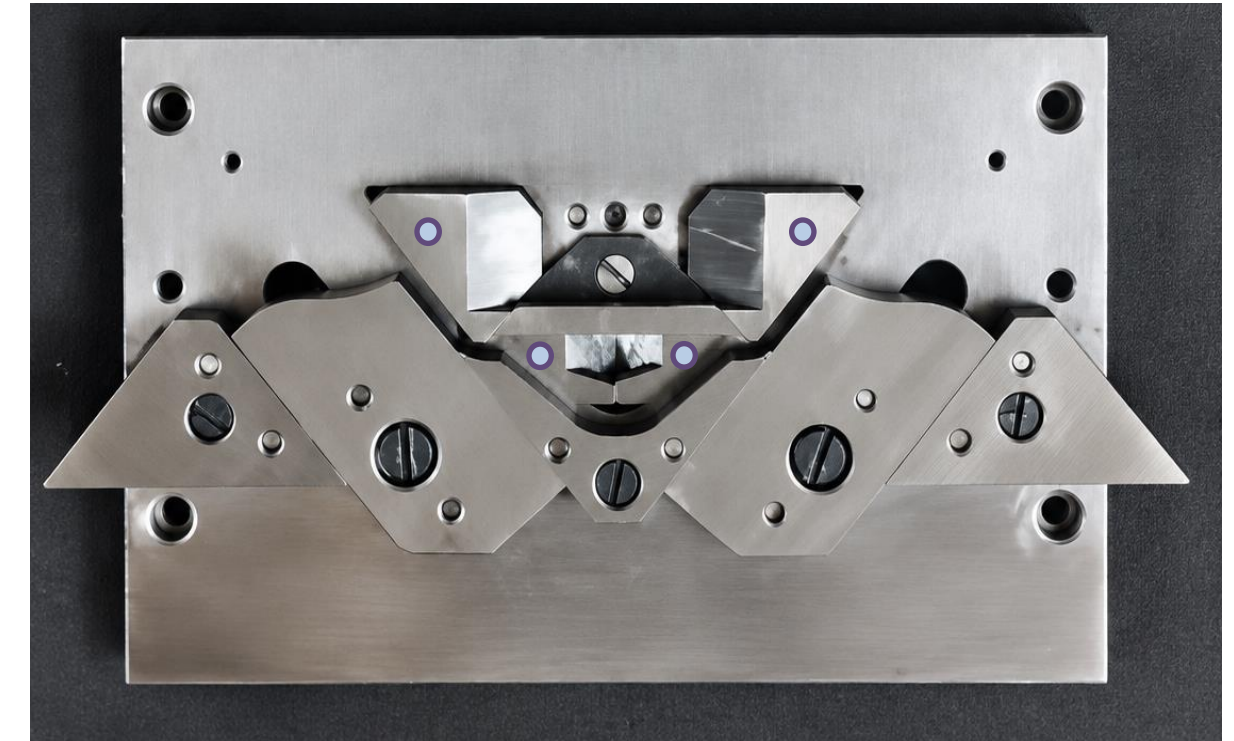
HALF CARDIGAN RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

❖ For 1st Course:

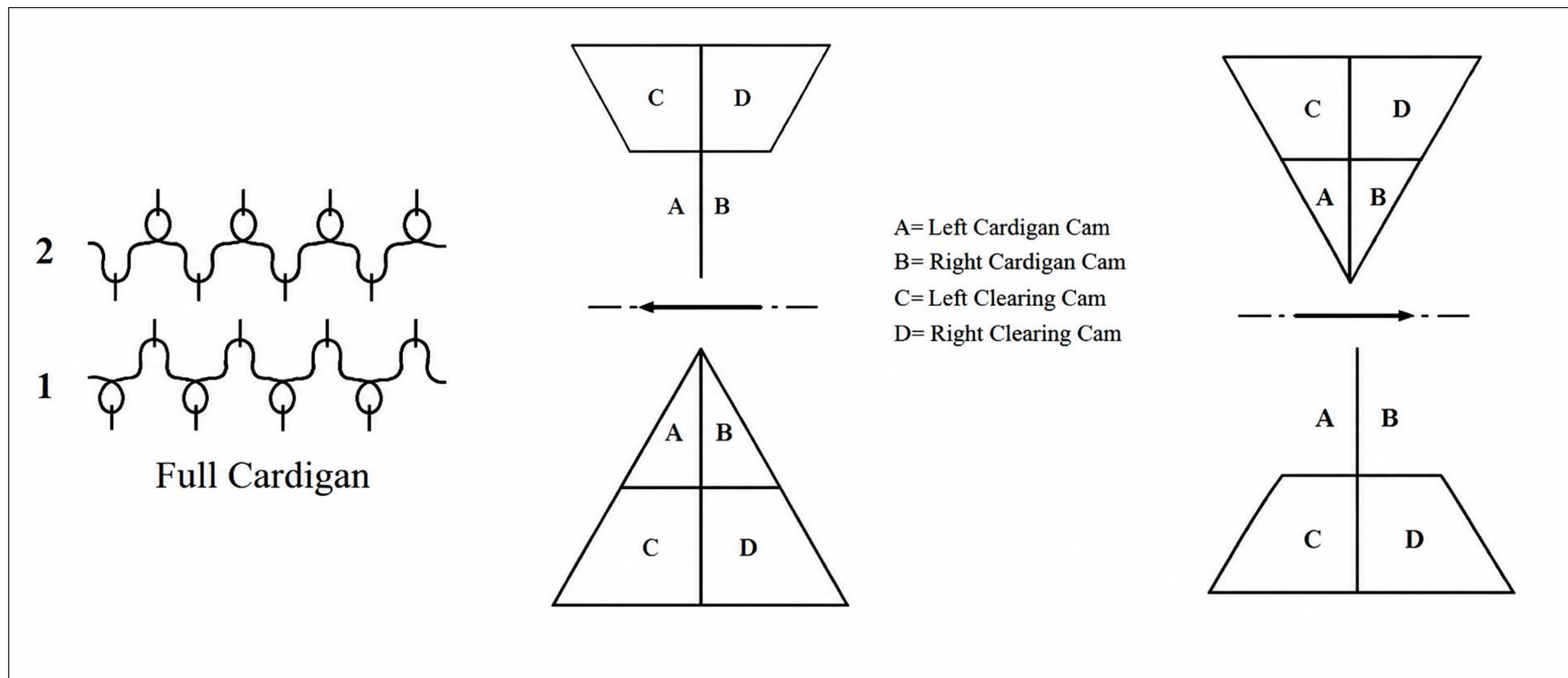
1. All cams at the front & back bed are in action.
2. When the carriage travel from *right to left*, all the needles will form knit loops like 1 x 1.

❖ For 2nd Course

1. Both cardigan cams at the *front bed (A & B) are out of action.*
2. All cams at the back bed are in action.
3. When the carriage will move from *left to right*, the *back bed needles will only form knit loops & front bed needles only form tuck loops.*



FULL CARDIGAN RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE



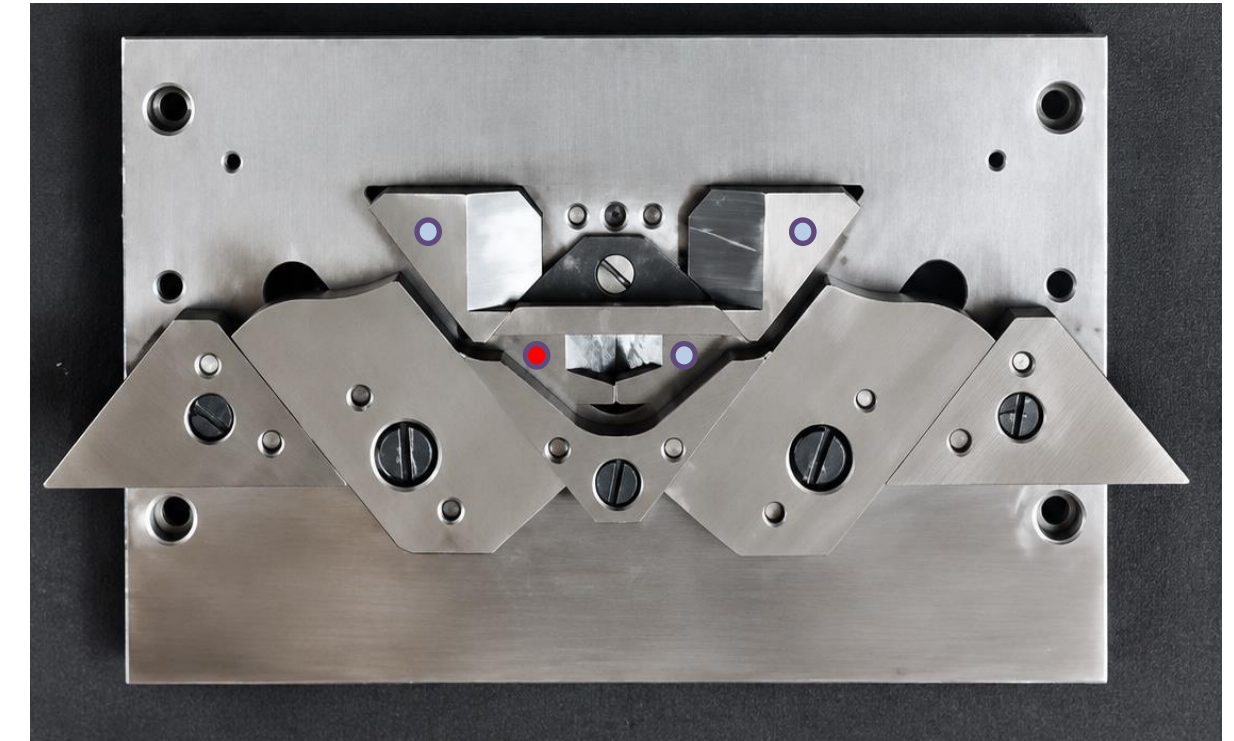
FULL CARDIGAN RIB FABRIC PRODUCTION IN FLAT KNITTING MACHINE

❖ For 1st Course:

1. Both cardigan cams at the *back bed (A & B)* are out of action.
2. All cams at the front bed are in action.
3. When the carriage will move from *right to left*, the *back bed needles will only form tuck loops & front bed needles only form knit loops.*

❖ For 2nd Course

1. Both cardigan cams at the *front bed (A & B)* are out of action.
2. All cams at the back bed are in action.
3. When the carriage will move from *left to right*, the *back bed needles will only form knit loops & front bed needles only form tuck loops.*



RACKING IN FLAT KNITTING MACHINE

Racking is the *sideways movement of one needle bed in relation to the other needle bed*.

This movement changes the relative position of the needles.

- ❖ Racking is measured by number of needle spaces.
- ❖ Needle beds can be moved in both directions (right or left).
- ❖ The needle bed can be racked by:
 - Half needle space
 - Full needle space.
- ❖ Racking is done by rotating a racking wheel.
- ❖ Each step of the wheel corresponds to the machine gauge.
- ❖ The movement can be:
 - Controlled manually
 - Controlled automatically.

RACKING IN FLAT KNITTING MACHINE

Purpose of Racking

Racking is mainly done for two reasons:

1. Loop Transfer

- Needle beds are moved so that the tricks become almost aligned.
- This allows needles from one bed to enter the transfer springs of the opposite bed.
- It helps in transferring loops from one needle bed to another.

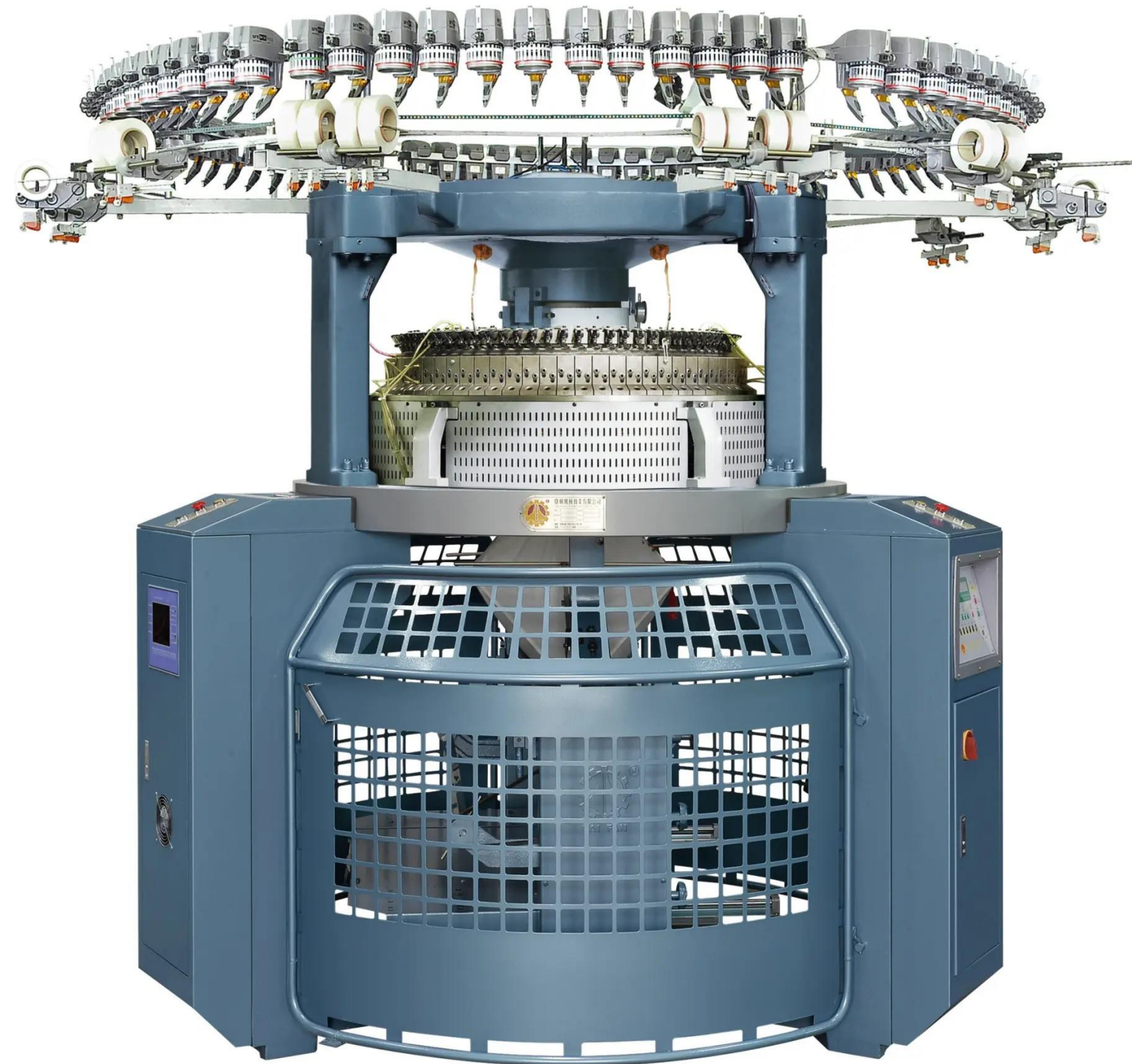
2. Pattern Formation

- One needle bed can be shifted a few needles sideways while still maintaining knitting position.
- Move transferred loops to a new position.
- Create inclined patterns in knitted fabric.

ADVANTAGES OF FLAT KNITTING MACHINE

- ❖ The flat machine is the *most versatile of weft knitting machines*, its stitch potential includes needle selection on one or both beds, racked stitches, needle-out designs, striping, tubular knitting, changes of knitting width and loop transfer.
- ❖ *A wide range of yarn counts may be knitted per machine gauge* including a number of ends of yarn in one knitting system, the stitch length range is wide and there is the possibility of changing the machine gauge.
- ❖ The operation and supervision of the machines of the simpler type is relatively *less arduous than for other weft knitting machines*.
- ❖ The *number of garments or panels simultaneously knitted* across the machine is dependent upon its knitting width, yarn carrier arrangement, yarn path and package accommodation.

CIRCULAR KNITTING MACHINE



FEATURES OF CIRCULAR KNITTING MACHINE

1. Circular knitting machine has normally rotating (clockwise) cylindrical needle bed.
2. On circular knitting machines mainly latch needles are used.
3. For single jersey machine holding down sinkers are used, one between every needle space. For double jersey machine, dial needle bed replaces the sinker.
4. Normally stationary angular cam systems are used for needle and sinker.
5. Latch needle cylinder and sinker ring or dial revolve through the stationary knitting cam systems.
6. For single jersey machine, sinker ring which is simply and directly attached to the outside top of the needle cylinder.
7. Stationary yarn feeders are situated at regular intervals around the circumference of the rotating cylinder.
8. Yarn is supplied from cones placed either on an integral over head bobbin stand or on a free-standing creel through tensioners, stop motions and yarn guide eyes down to the yarn feeder guides.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

There are *three main types* of circular knitting machines:

A) Revolving Cylinder Latch Needle Machines

These machines produce most weft knitted fabrics.

They have two main types:

1. Open Top / Sinker Top / Single Jersey Machine

- ☐ It has one set of needles arranged in the cylinder.
- ☐ It is mainly used to produce single jersey fabrics.
- ☐ It may have pattern-making ability or may not.

2. Dial and Cylinder Machine

- ☐ It has two sets of needles: Cylinder needles (vertical) & Dial needles (horizontal)
- ☐ These machines produce: Rib fabrics, Interlock fabrics.
- ☐ They may have patterning systems.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

B) Revolving Cylinder Bearded Needle Single Jersey Machine

There are two types:

1. Sinker Wheel Machine
2. Loop Wheel Frame

Both types have similar features:

- Needles are fixed in the needle bed.
- The needle bed rotates during knitting.
- Special parts move the yarn and loops along the needles.
- The fabric is produced in a tube shape.
- The technical back side of the fabric faces outside.
- They use fewer feeders.
- Their production speed is lower, but they can make: Special knitted structures. High-quality fabrics.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

C) Circular Garment Length Machines

- ☐ These machines are used to produce complete garment lengths (large pieces of knitted fabric).
- ☐ They are usually body-width size or larger.
- ☐ They may have: Cylinder and dial arrangement, or Double cylinder arrangement.
- ☐ Small diameter versions are used for hosiery products (such as socks).
- ☐ Hosiery machines may have:
- ☐ Single cylinder, Cylinder and dial, or Double cylinder.

FABRIC LENGTH MACHINE



FEATURES OF FABRIC LENGTH MACHINE

- Fabric machines are circular knitting machines that produce *continuous tubular fabric of fixed width*.
- They are *large diameter latch needle machines* that work at *high speed*.
- After the required length is knitted, the fabric is removed manually, usually in roll form.
- Most fabrics are produced on single cylinder (single jersey) and cylinder-dial (double jersey) machines because they have high production speed and efficiency.
- *Tubular fabric may need to be opened into flat width fabric by splitting, unless a tubular shape is required.*
- After knitting, the fabric is finished and cut into garments or used for household and technical purposes.
- Fabric machines differ in productivity, flexibility, and pattern-making ability.
- Usually, cam settings and needle arrangements are not changed during fabric production.

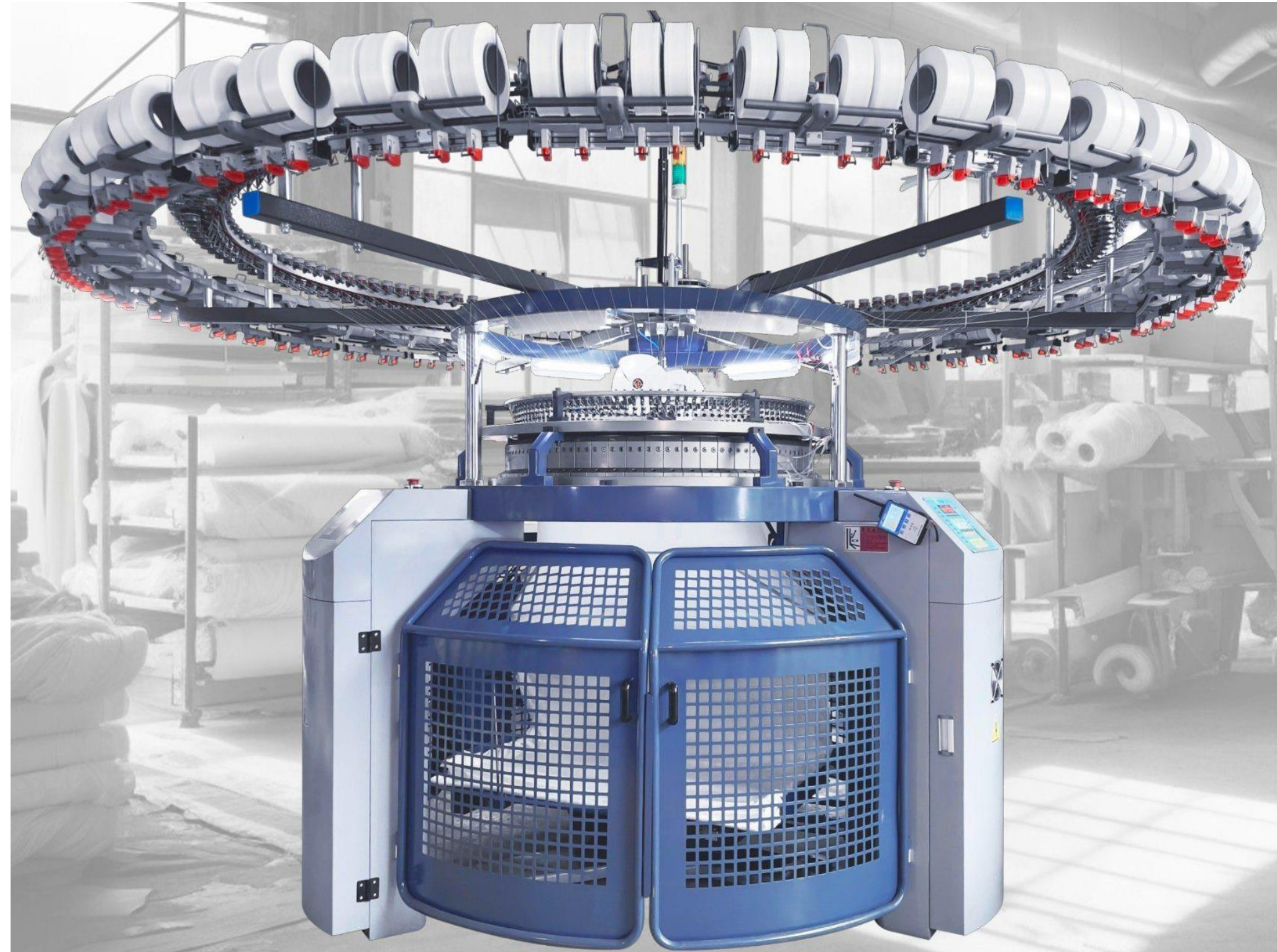
GARMENTS LENGTH MACHINE



FEATURES OF GARMENTS LENGTH MACHINE

- Garment-length machines include *flat machines, hosiery machines, glove machines, straight bar frames, and circular garment machines.*
- They *produce complete garment-length pieces instead of continuous fabric.*
- These machines use a control system with timing and counting devices to manage the garment knitting sequence.
- The garment width may change during knitting according to the design.
- They *generally have a coarser gauge than fabric machines.*
- The take-down system is more advanced because it must handle different production speeds and finished garment removal.
- Garments can be knitted: In tubular form, or In open width form.
- They are used to produce: Knitwear (sweaters, pullovers, cardigans), Outerwear, Underwear.

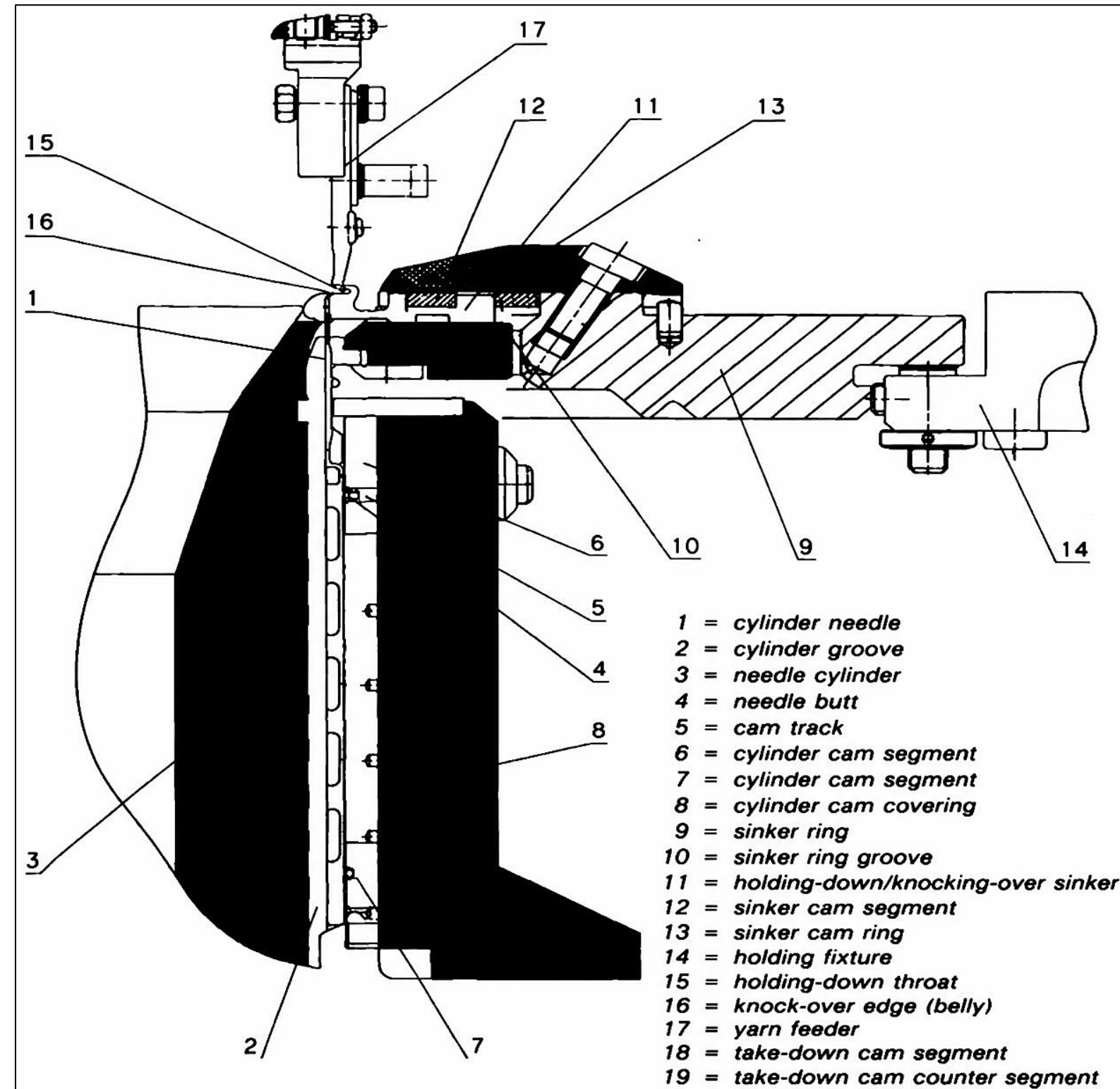
SINGLE JERSEY CIRCULAR KNITTING MACHINE



FEATURES OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

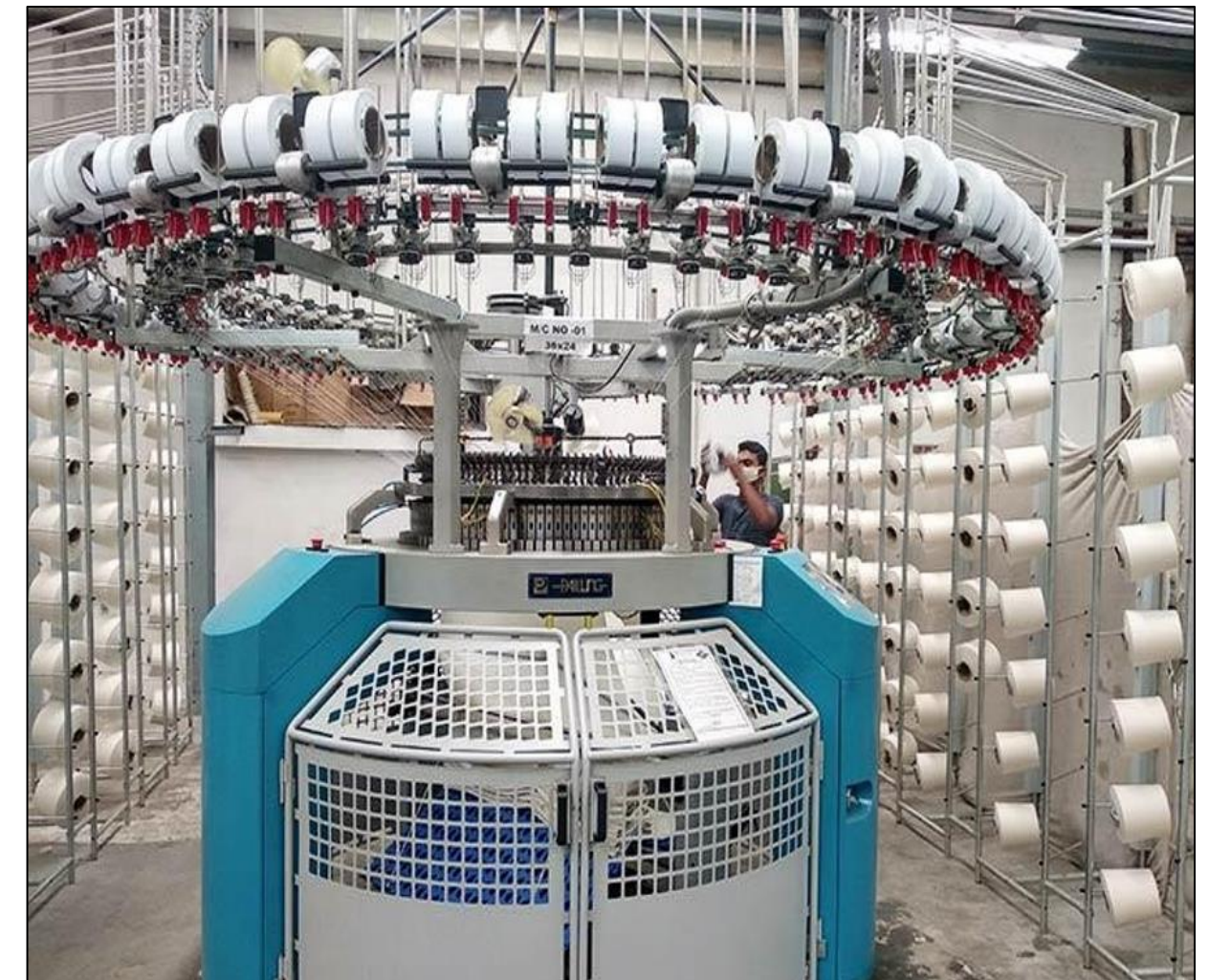
1. It is a single-bed circular knitting machine having only one set of needles.
2. The needles are latch-type and mounted vertically in the slots of the needle cylinder.
3. The sinkers are fitted between every two needles and move radially.
4. The machine produces plain fabric (single jersey) which has technical face and back sides.
5. Usually, the diameter of the machine ranges from 12 to 60 inches.
6. The gauge may vary from 14 to 36 needles per inch, depending on yarn count and end use.
7. Cam systems are placed around the cylinder to control needle movement for knit, tuck, and miss stitches.
8. The yarn feeders (generally 90-120 per machine) feed yarns continuously to each knitting head.
9. Sinker ring which is simply and directly attached to the outside top of the needle cylinder.
10. The fabric is drawn down through the center of the cylinder by take-down rollers.
11. The production rate depends on machine diameter, speed (revolutions per minute - RPM), and number of feeders.
12. The machine frame holds the main drive, take-down mechanism, fabric collection, and lubrication systems.
13. Fabrics produced are used for T-shirts, underwear, sportswear, and other lightweight garments.

SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS



SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

- Creel
- Cone package holder
- Yarn passing tube
- Tensioner
- Positive feeder
- VDQ pully
- Stop motion device
- Feeder
- Needle
- Cam
- Cam box
- Cylinder
- Sinker
- Fabric spreader
- Take down roller
- Cloth roller
- Lubrication system
- Dust removing fan

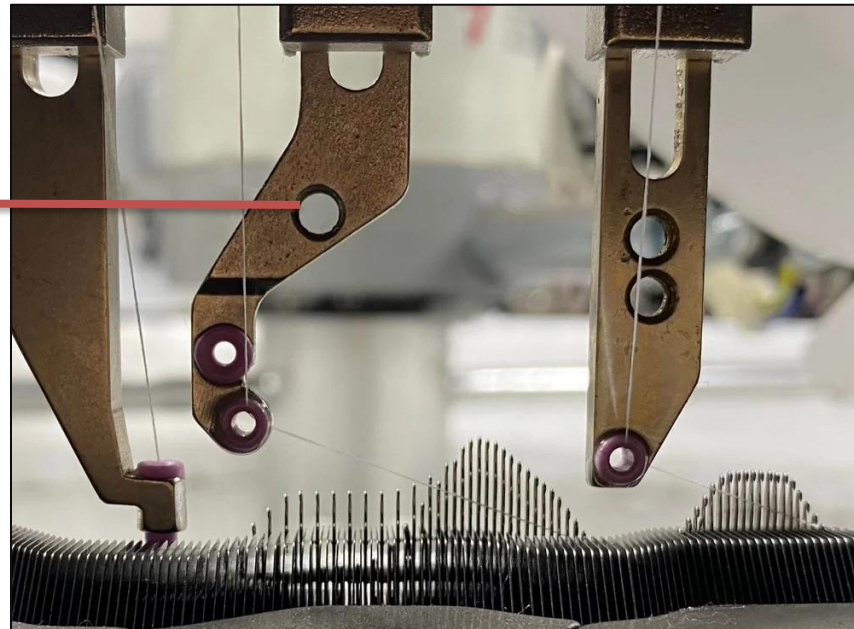


SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

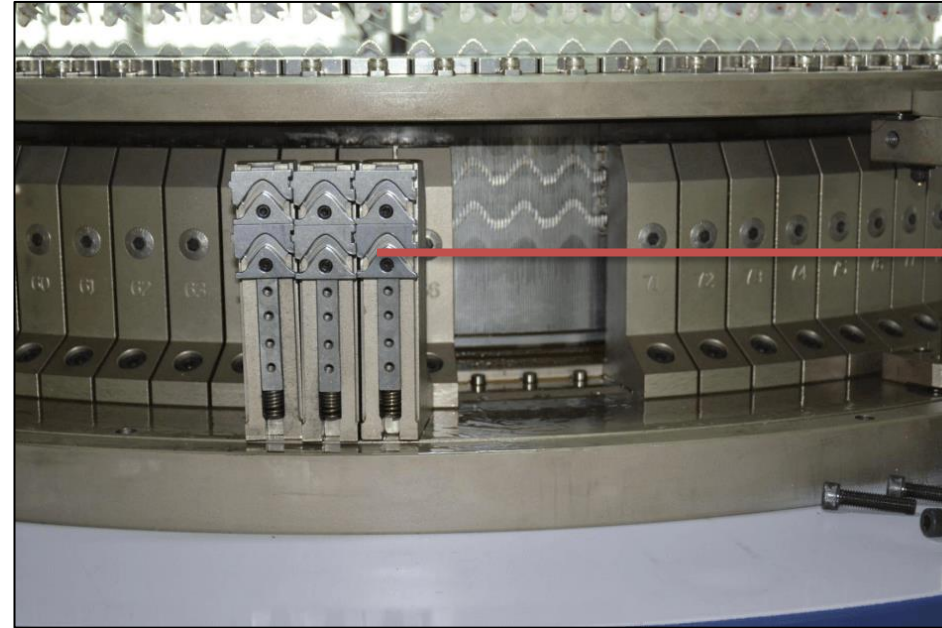


SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

YARN FEEDER



CAM



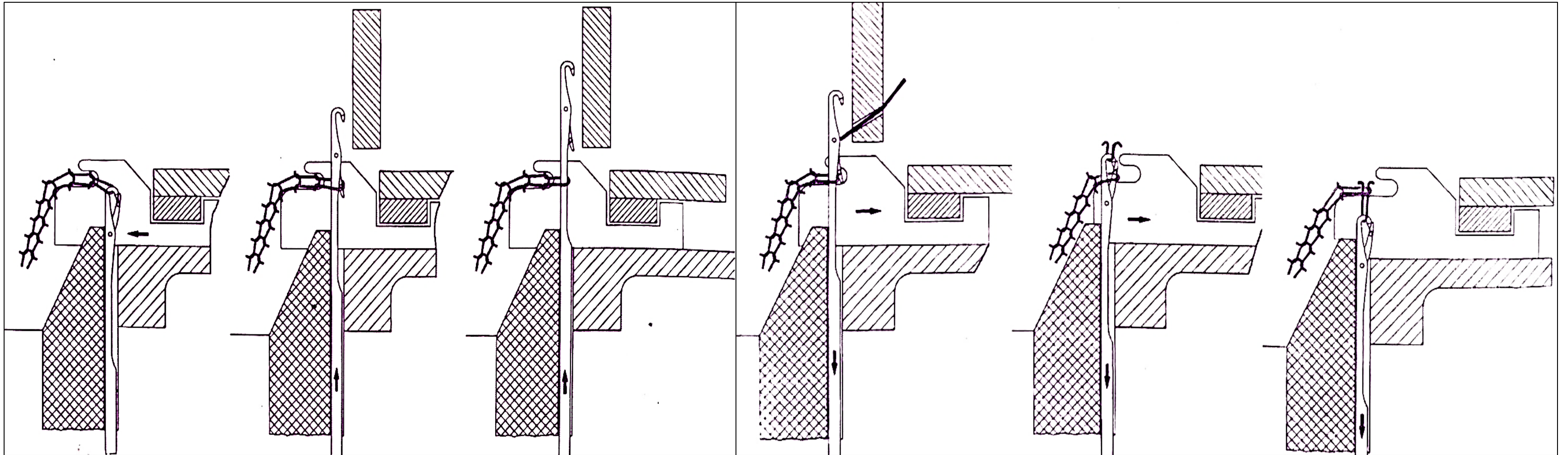
CYLINDER



SPREADER



KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE



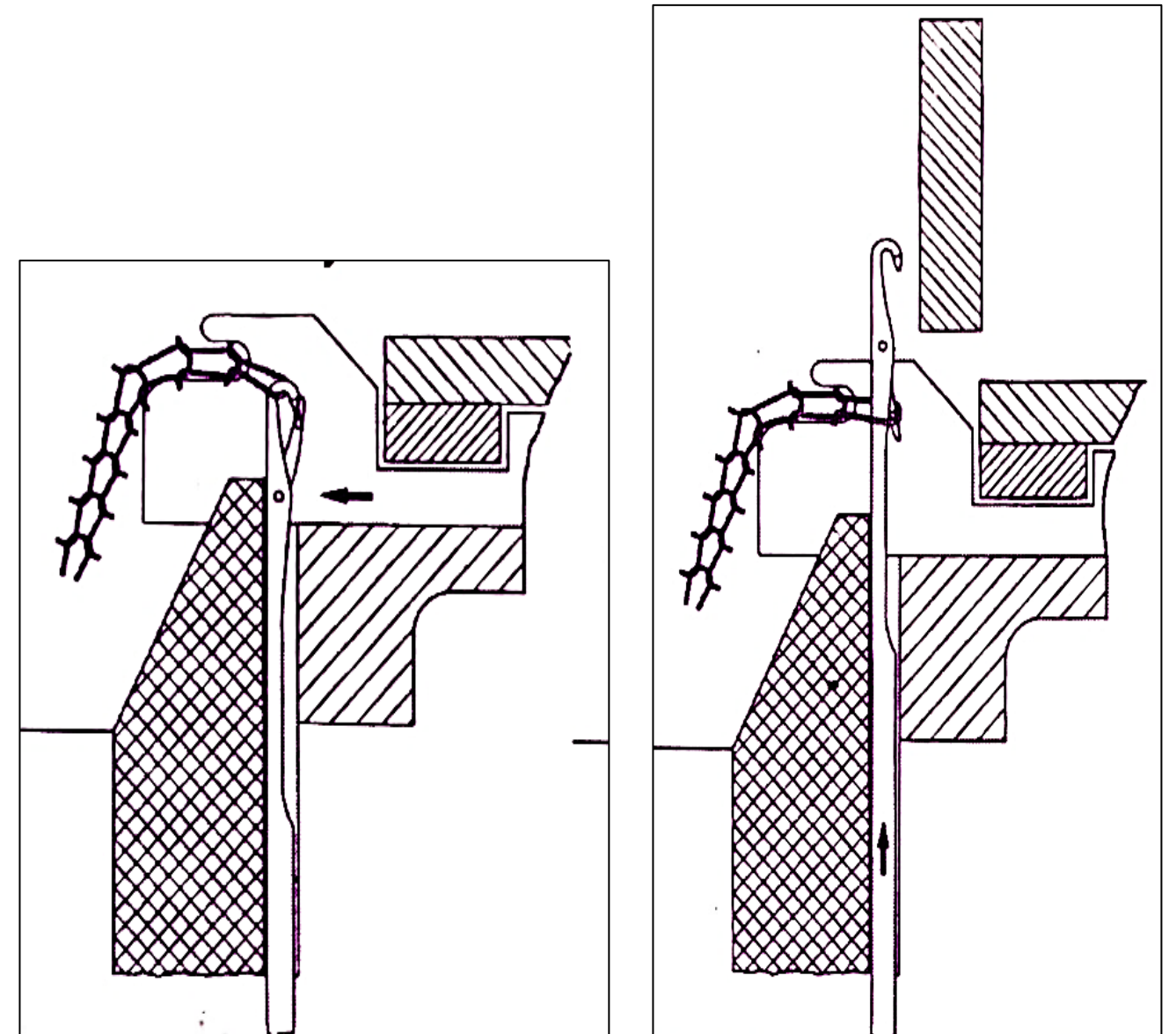
KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

Position 1: Rest-in Position

In the rest position, the *needle is at its starting point*. The top edge of the needle head is *level with the knock-over edge of the sinker*. The *sinker moves towards the center of the cylinder and its throat holds the old loop* and keeps the fabric in position.

Position 2: Tucking-in Position

In this position, the *needle moves upward from the rest position*. During this movement, the *old loop opens the needle latch and rests on the latch*. The yarn feeder keeps the latch open and prevents it from closing. *The sinker does not move during this stage*.



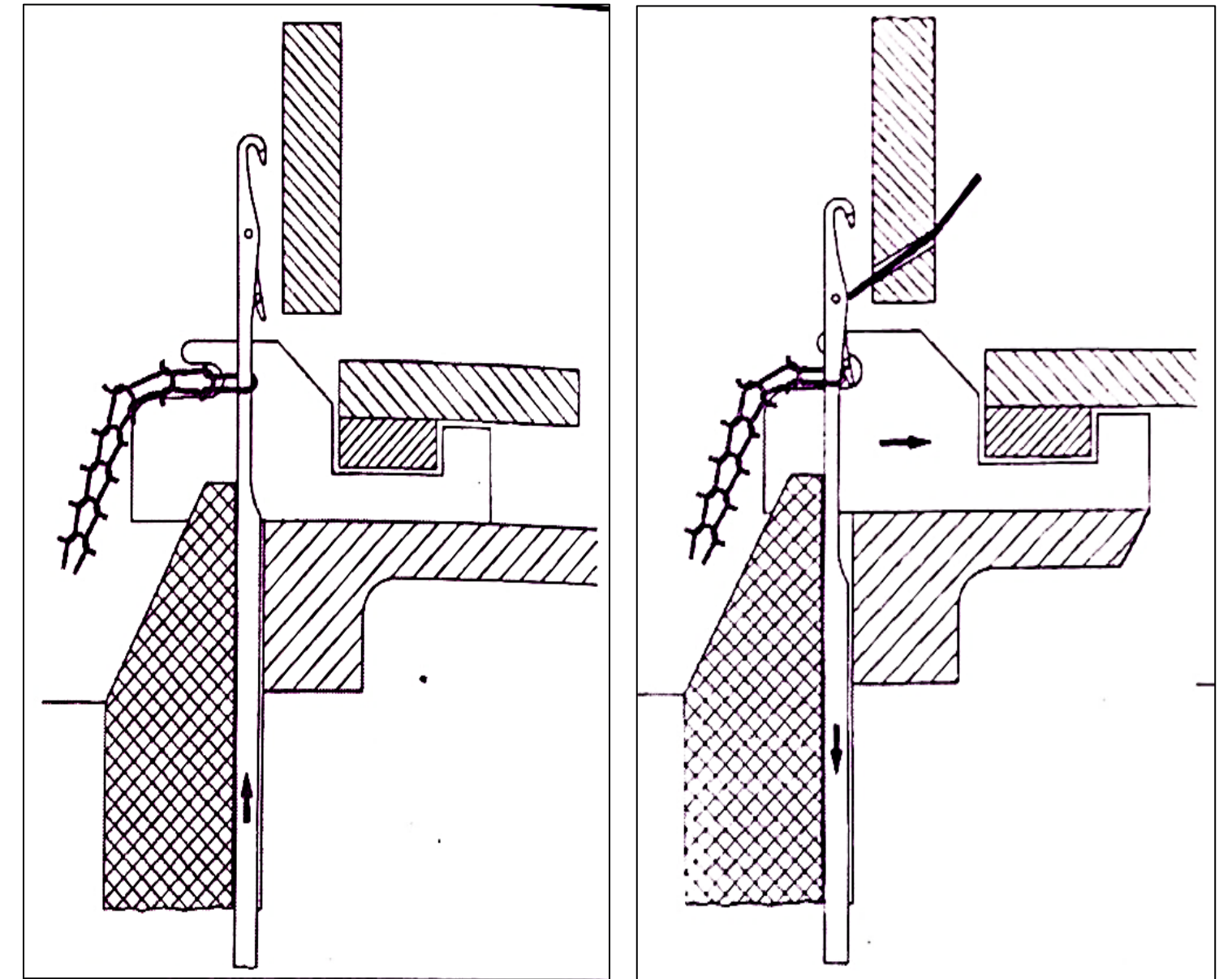
KNITTING ACTION OF SJCKM

Position 3: Clearing Position

The needle moves further upward and reaches its highest position. *Since the sinker holds the fabric down, the old loop slides along the needle stem and moves below the latch.* The yarn feeder prevents the latch from closing, and the sinker remains still.

Position 4: Yarn Presenting Position

The *needle starts moving downward from the clearing position.* Before the old loop closes the latch, *the yarn feeder places the new yarn inside the needle hook.* After the yarn is fed, *the latch can close during further downward movement.* At this stage, *the sinker moves away from the cylinder center because the fabric no longer needs to be held by the sinker throat.*



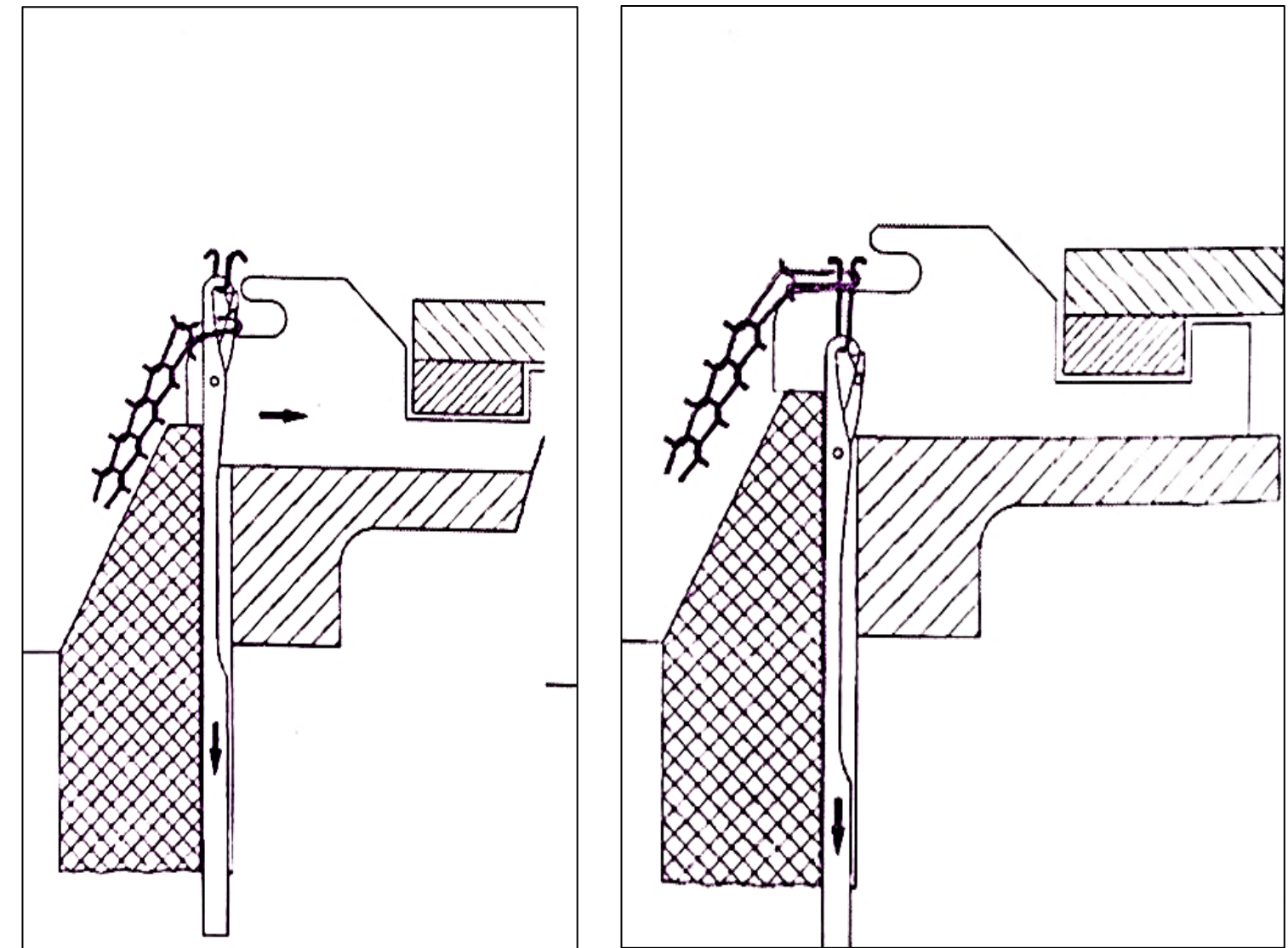
KNITTING ACTION OF SJCKM

Position 5: Landing Position

The needle continues moving downward from the yarn presenting position. The old loop closes the needle latch, and *the new yarn is trapped inside the closed needle hook*. The sinker reaches its extreme position and *holds the old loop at the knock-over edge so that the new yarn can be pulled through it*.

Position 6: Knocking over Position

The needle moves further downward and pulls the new yarn through the old loop. *The old loop slips off the needle and a new stitch is formed*. The movement of the take-down and guide segments controls the length of the newly formed loop, allowing accurate adjustment of stitch size.



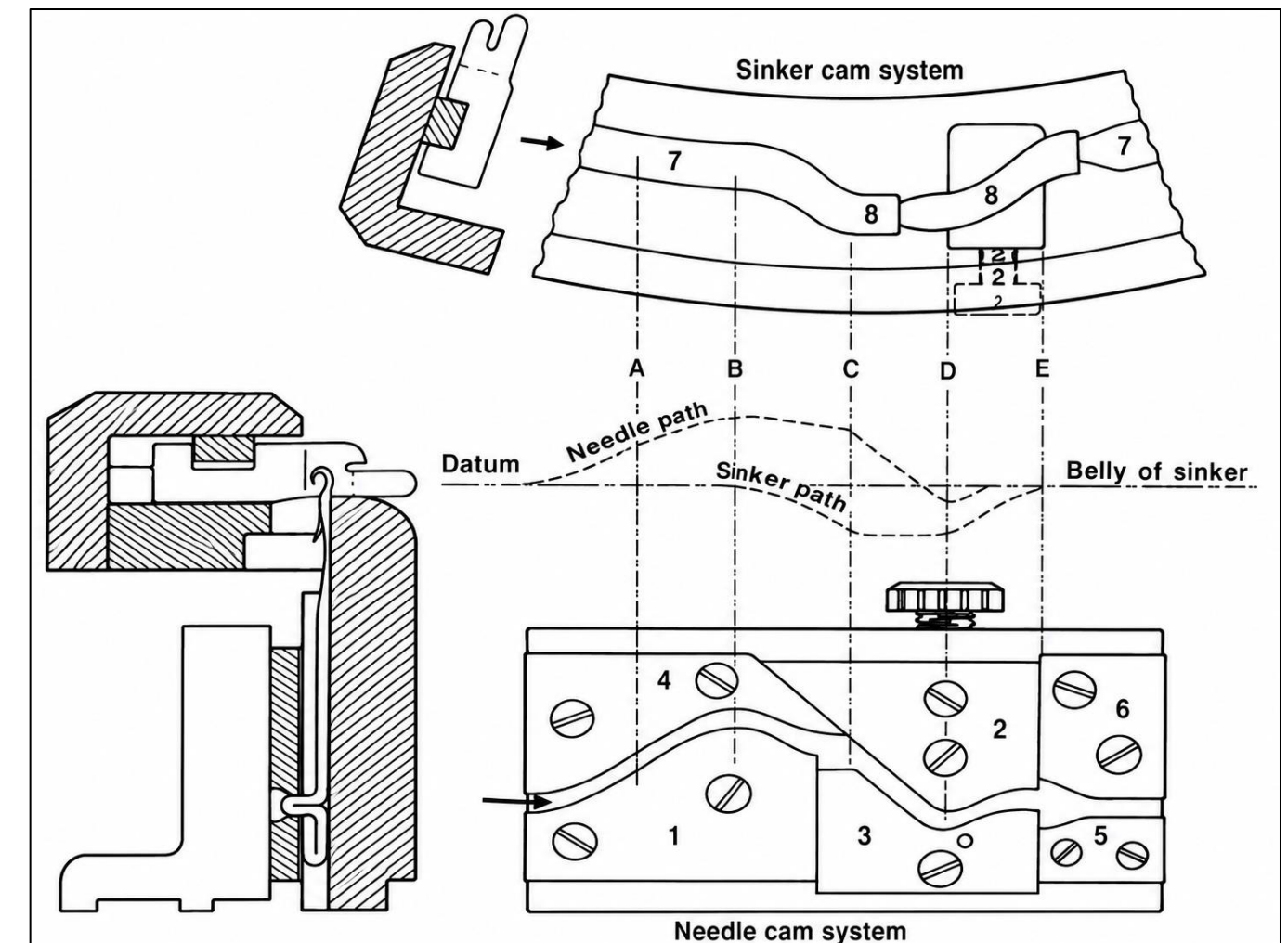
CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The cam system has two main parts: *Needle cam system & Sinker cam system.*

The needle cam race consists of six main cams:

1. Clearing cam (Raising cam)
2. Stitch cam (Lowering cam)
3. Upthrow cam.
4. Guard cam of clearing cam.
5. Return cam brings the needle back to its position.
6. Guard cam of return cam.

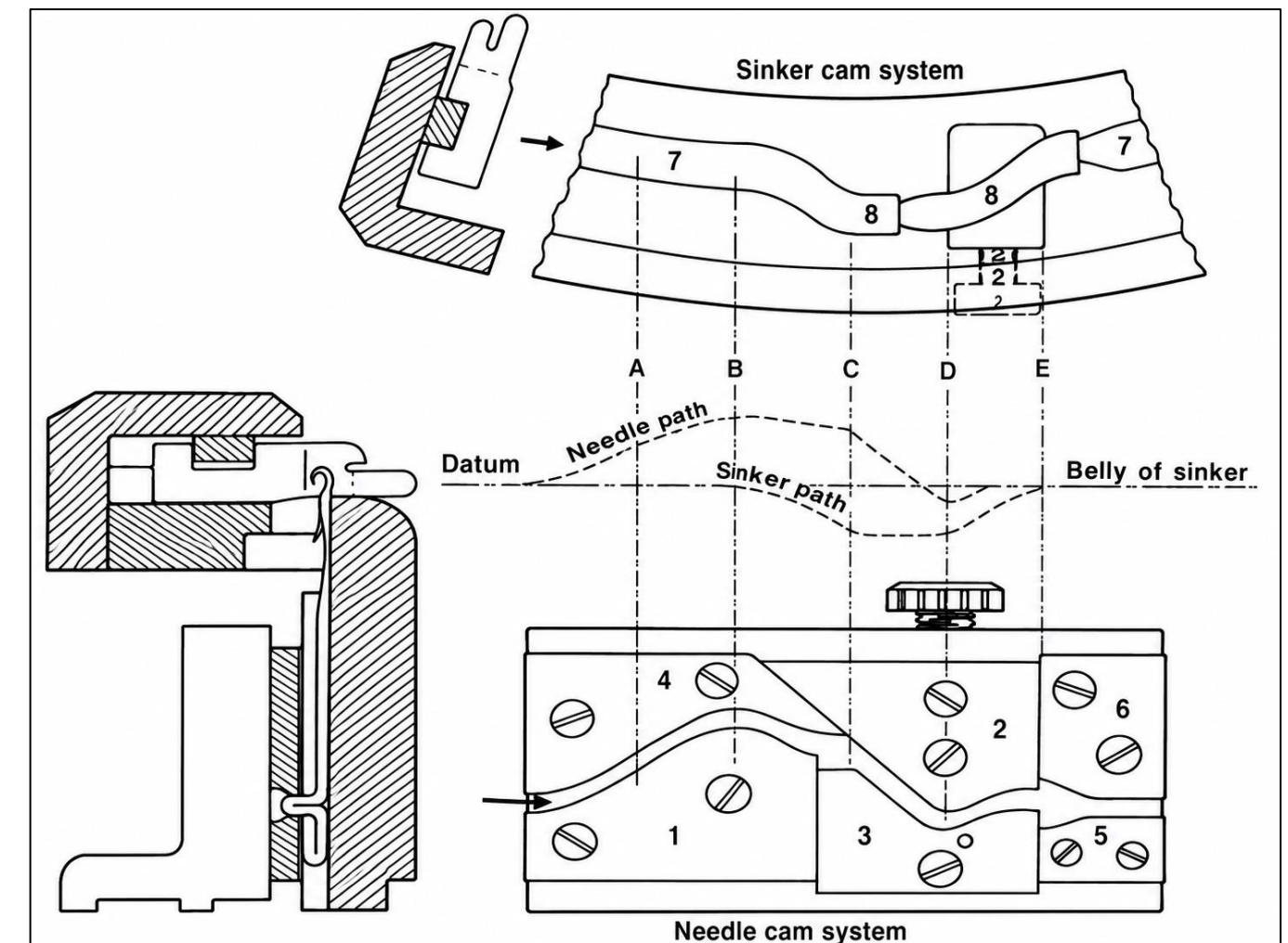
The stitch cam (2) and upthrow cam (3) can be adjusted vertically to change the stitch length.



CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The sinker cam race consists of the followings:

- 7. The race cam
- 8. The sinker- withdrawing cam and
- 9. The sinker- return cam.

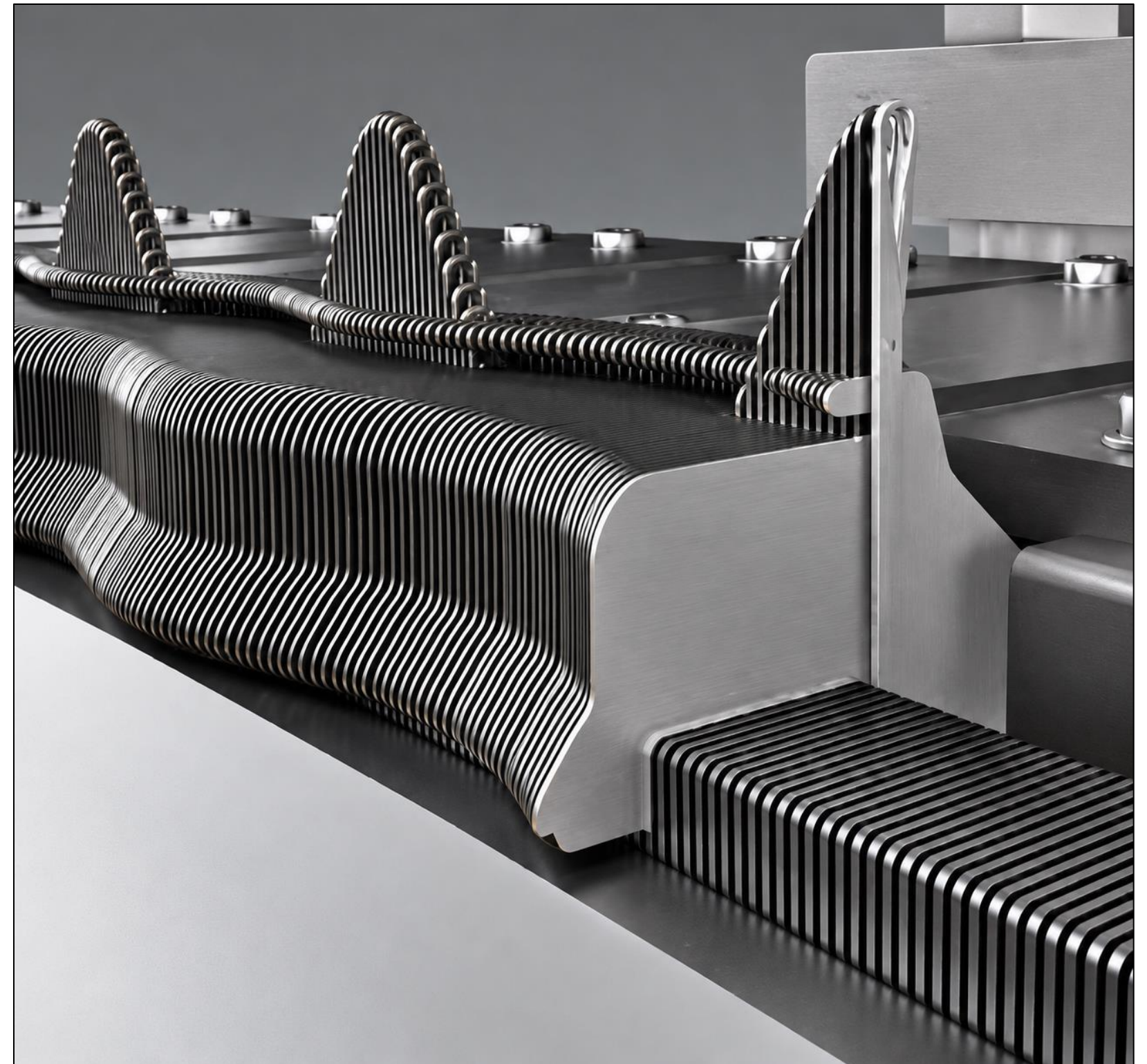


SINKER TIMING

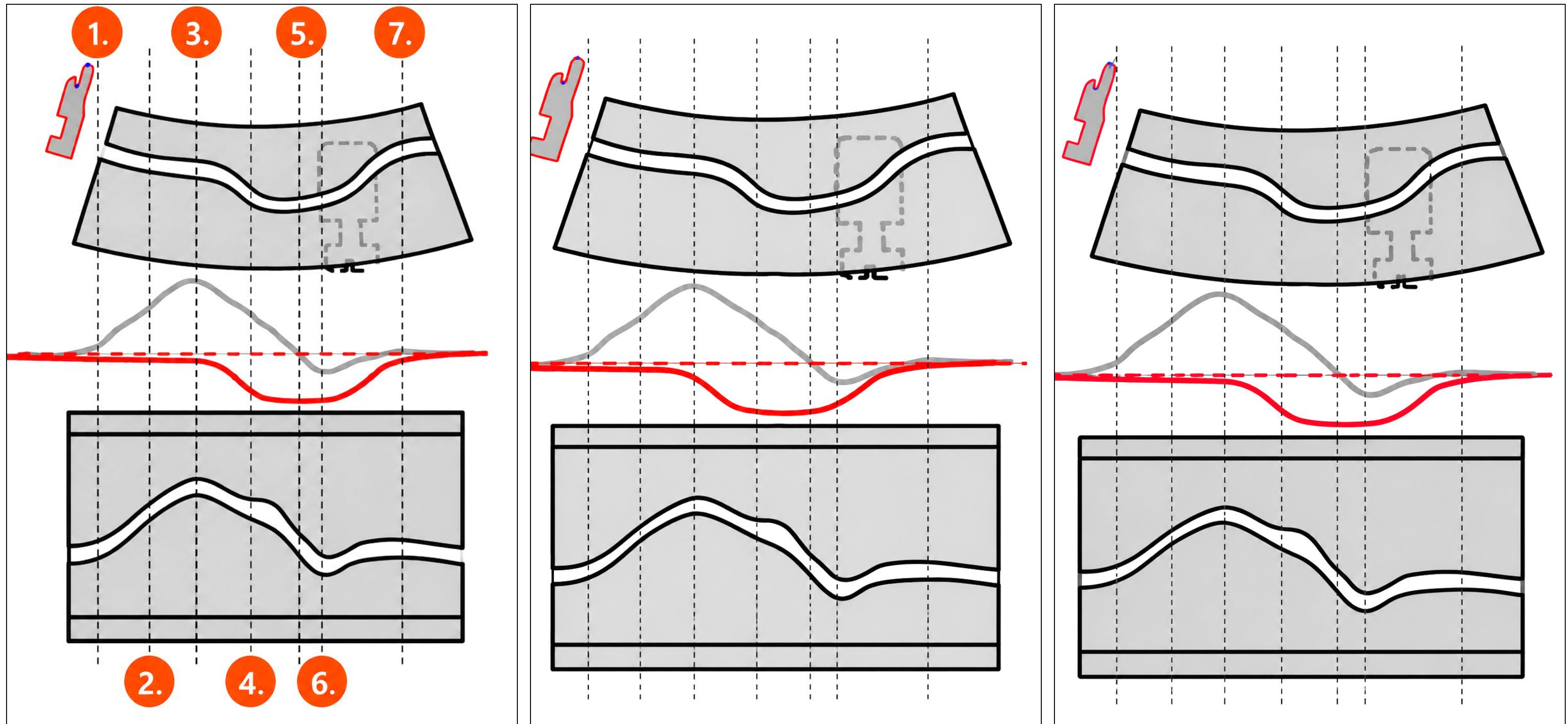
While the machine runs, the sinker performs two types of movement, *an inward and outward movement* and *a radial movement*. The sinker cam ring remains stationary, but the *sinker ring along with the cylinder moves with same rpm* to avoid needle sinker collision. *The most forward (inward) position of the sinker during the knitting cycle is known as the Push Point and its relationship to the needle's movement is known as the Sinker Timing.*

3 Types of Sinker Timing:

1. Synchronized Timing
2. Advanced Timing
3. Delayed Timing.



SINKER TIMING



SINKER TIMING

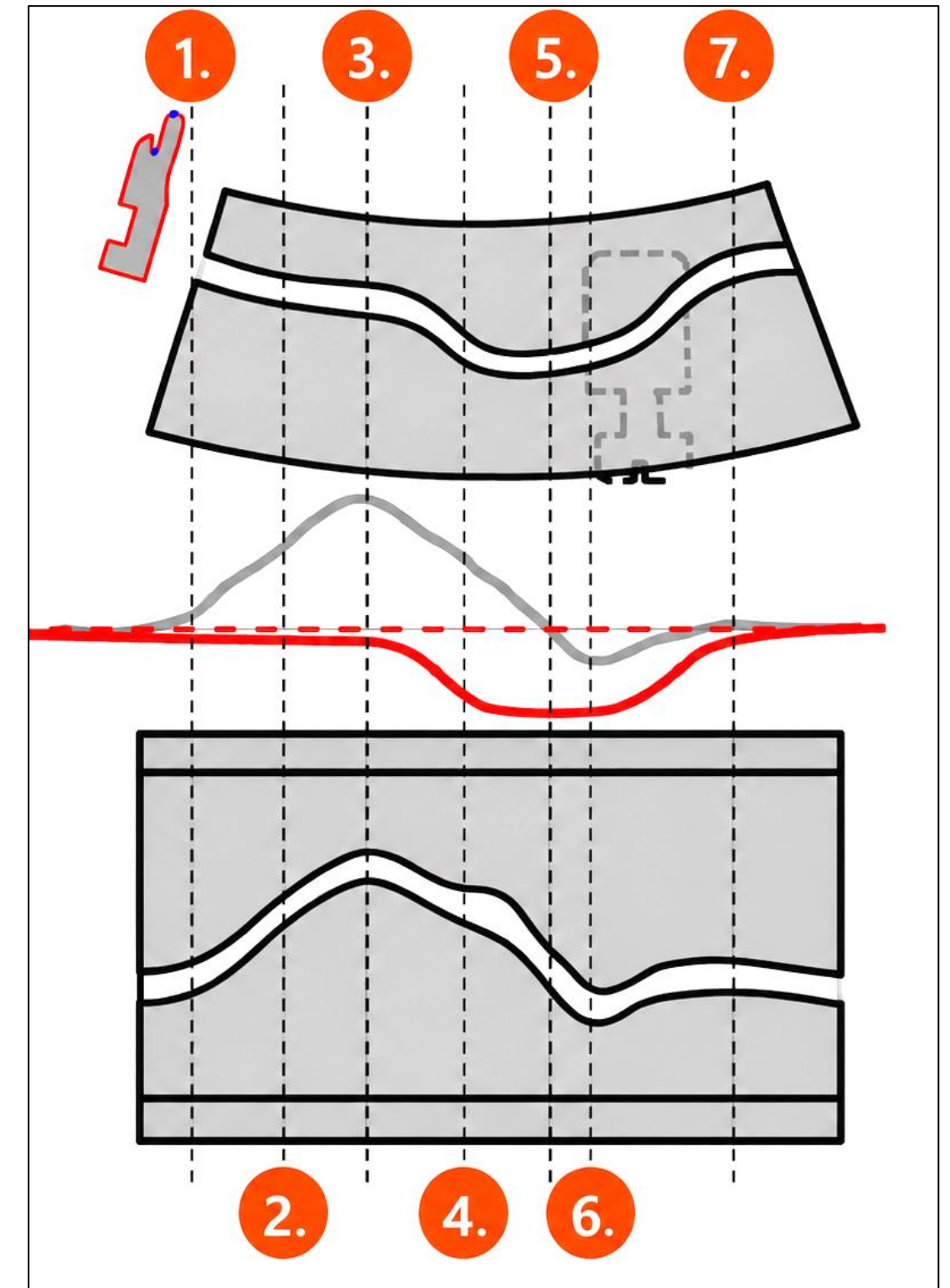
Synchronized Timing:

Sinker starts its inward movement after couliering.

Effects of Synchronized Timing:

- ❑ Balanced structure.
- ❑ Sinker loop and needle loops are equal size.

- | | |
|-----------------------------------|------------------------------------|
| ❖ <i>Position 1: Rest-in</i> | ❖ <i>Position 5: Knocking over</i> |
| ❖ <i>Position 2: Tucking in</i> | ❖ <i>Position 6: Couliering</i> |
| ❖ <i>Position 3: Clearing</i> | ❖ <i>Position 7: Rest-in</i> |
| ❖ <i>Position 4: Yarn Feeding</i> | |



SINKER TIMING

Advanced Timing:

Sinker starts its inward movement before finishing the couliering.

Effects of Advanced Timing:

- ❑ Stress in yarn is high.
- ❑ Sinker loop is larger than needle loop.
- ❑ Unbalanced fabric structure.

❖ *Position 1: Rest-in*

❖ *Position 2: Tucking in*

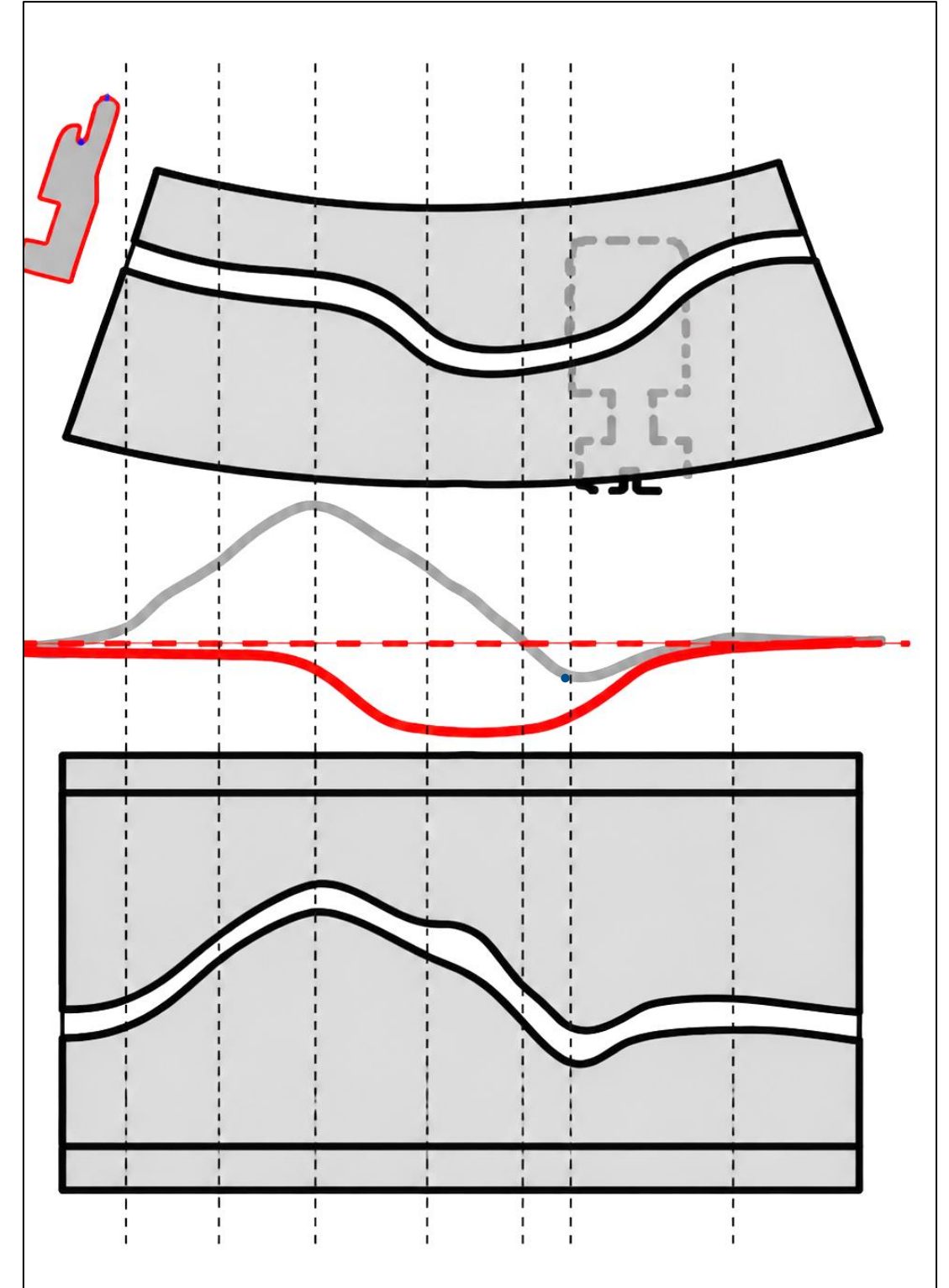
❖ *Position 3: Clearing*

❖ *Position 4: Yarn Feeding*

❖ *Position 5: Knocking over*

❖ *Position 6: Couliering*

❖ *Position 7: Rest-in*



SINKER TIMING

Delayed Timing:

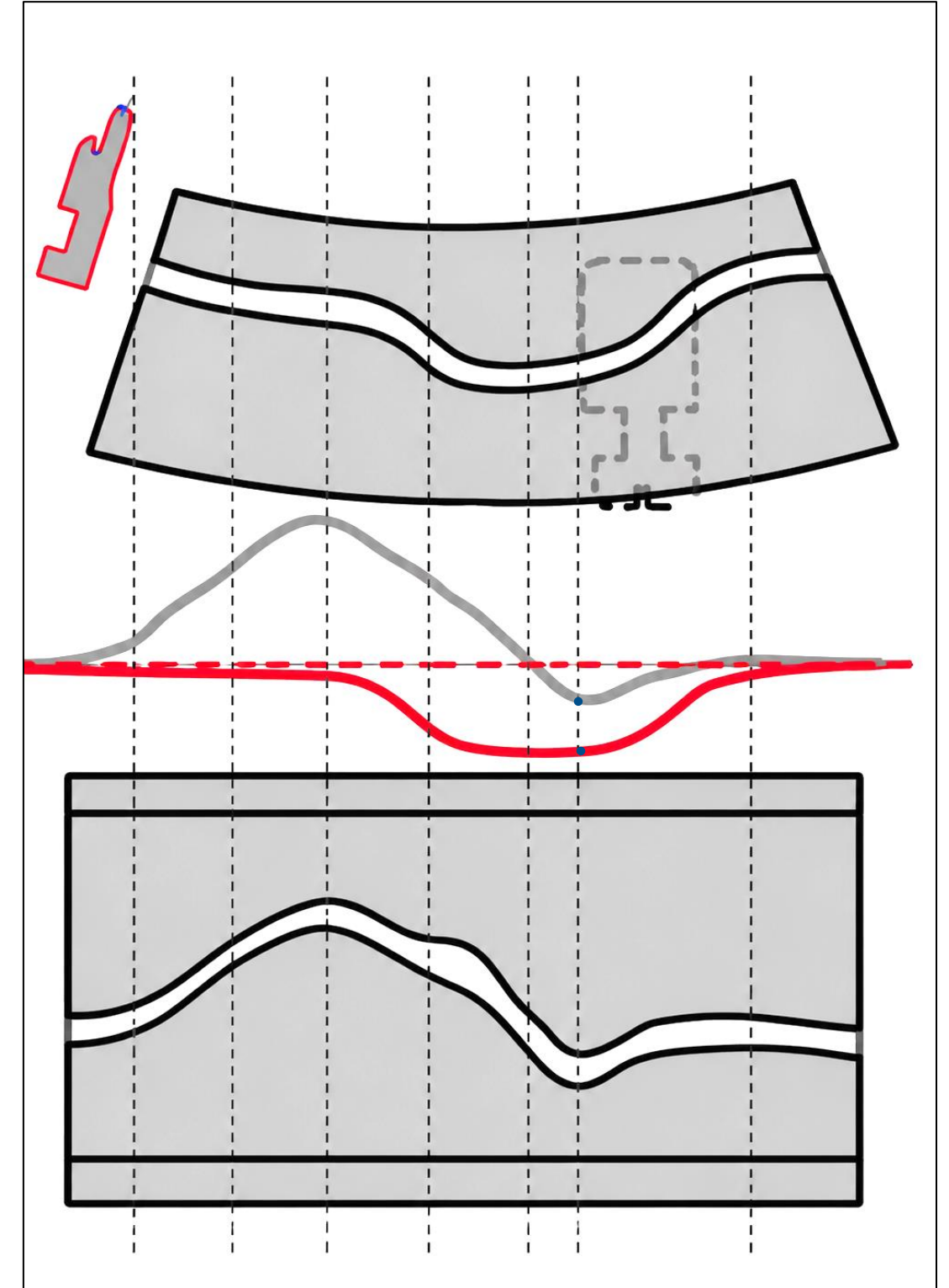
Sinker starts its inward movement after a while of finishing couliering.

Effects of Delayed Timing:

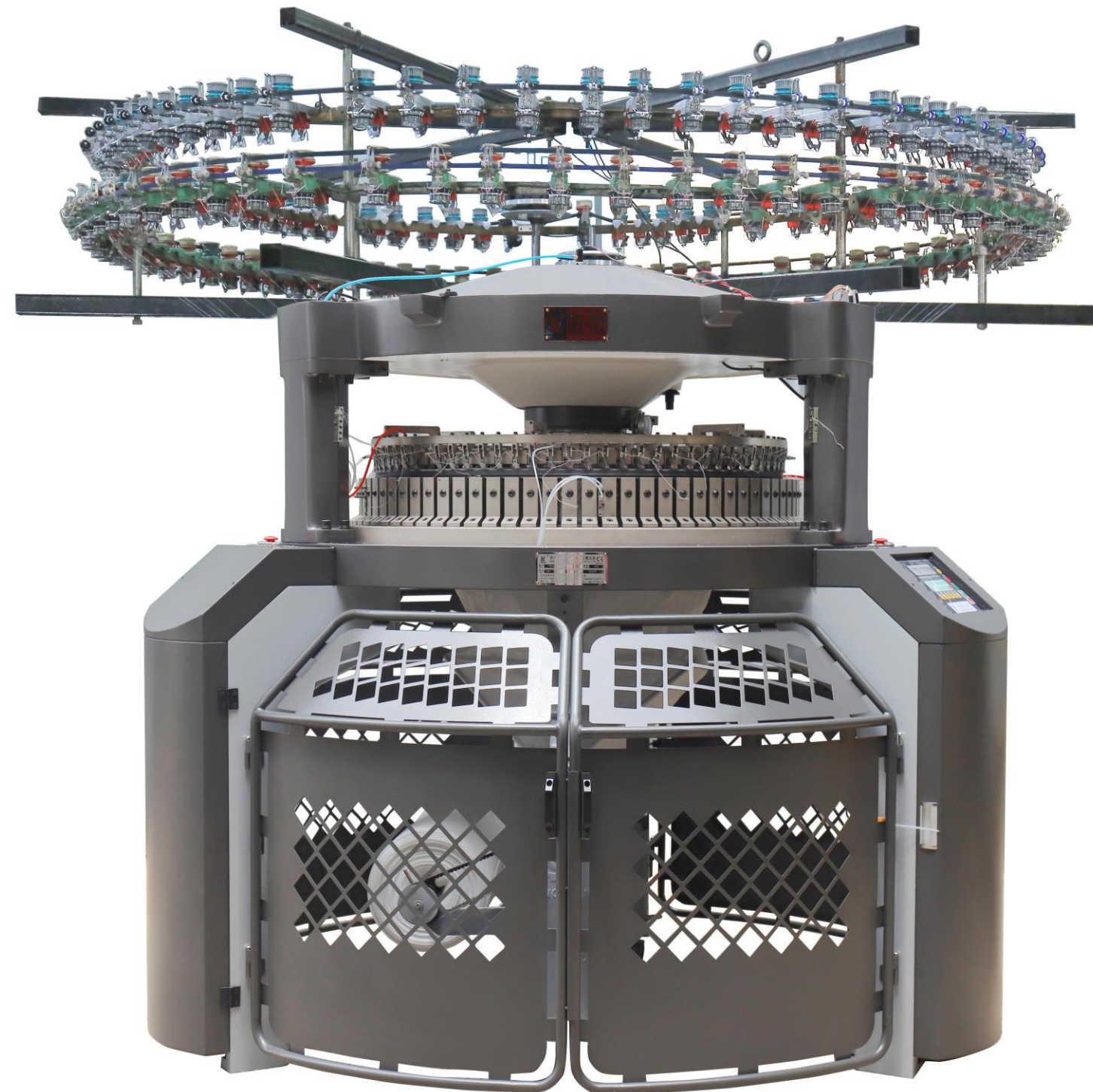
- ❑ Lower stress is induced on yarn loop.
- ❑ Needle loop is larger than sinker loop.

- ❖ *Position 1: Rest-in*
- ❖ *Position 2: Tucking in*
- ❖ *Position 3: Clearing*
- ❖ *Position 4: Yarn Feeding*

- ❖ *Position 5: Knocking over*
- ❖ *Position 6: Couliering*
- ❖ *Position 7: Rest-in*



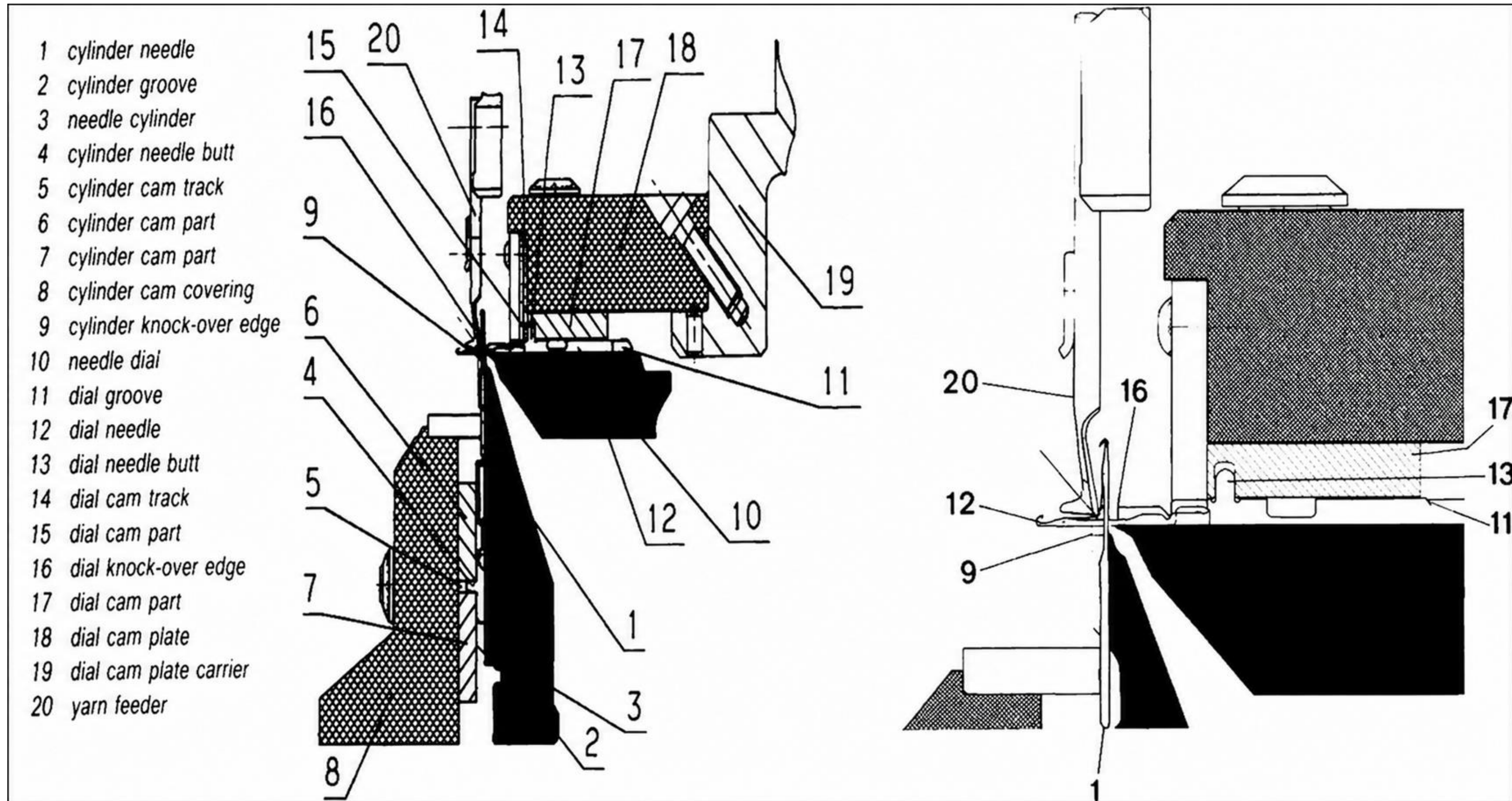
DOUBLE JERSEY CIRCULAR KNITTING MACHINE



DOUBLE JERSEY CIRCULAR KNITTING MACHINE

- ❖ In this machine group there is *one set of needles on the circumference of a vertical cylinder* and a *second set of needles, arranged perpendicular to the first set and mounted on a horizontal dial*.
- ❖ On most of the circular knitting machines the **cylinder and dial rotate**, whereas the **cams with yarn feeder guides are stationary**. The following figure shows a cross-sectional view of the region containing the knitting elements of a rib circular knitting machine. The set-up of the cylinder (3) with its knitting elements (1 to 9) is the same as with plain circular knitting machines. In a **horizontal dial** (10) grooves (11) are milled in. The latch needles (12) are housed and guided in these grooves. The dial needle (12) obtains its motion for stitch formation through its butt 13, which extends into a cam track (14). This cam track (14) is formed by the cam parts (15) and (17), which in turn are fixed to a dial cam plate (18). During the rotation of the cylinder and the dial the cylinder needle (1) is moved vertically and the dial needle 12 is moved horizontally, corresponding to the shape of the cam track in the cylinder and dial cams.
- ❖ In a gauge range from **5 to 20 needle per inch**.

DOUBLE JERSEY CIRCULAR KNITTING MACHINE



FEATURES OF RIB CIRCULAR KNITTING MACHINE

- ❖ Machine has two beds- may be flat or circular.
- ❖ There are two sets of needles- one in each bed.
- ❖ There are two cam systems- one in each bed.
- ❖ Needles in the two beds are not face to face but needles in one bed are in between the needles of the other bed so that they do not touch while raised for clearing.
- ❖ Rib fabrics are double-faced structures as well as balanced structures.
- ❖ Both face loops and back loops are visible on both sides of the fabric and fabric has an identical appearance in face and back.
- ❖ Each course is made of the face loop and back loop in an alternative order.
- ❖ Face loops are made by the needles in cylinder bed and back loops are made by needles in dial bed.

FEATURES OF RIB CIRCULAR KNITTING MACHINE

- ❖ Fabric is much thicker, generally double than single jersey fabric.
- ❖ Fabric does not curl at the free edges.
- ❖ Fabric can easily be unroved from the end last knitted.
- ❖ Loop length can be changed by varying dial height or vertical gap between two beds and timing of knitting in addition to stitch cam setting in both the beds.
- ❖ Rib machines require finer yarn and result in comparatively costly fabric.

NEEDLE GAITING

The way of two beds of knitting needles and their housings are *aligned with each other in a dial and cylinder* of circular knitting machine or flat bed knitting machine.

Generally, two types of needle gaiting are used. These are:-

- Rib gaiting
- Interlock gaiting

NEEDLE GAITING

1. Rib Gaiting: When the needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed **at alternate position** with each other is known as Rib gaiting.



Fig: Rib Gaiting

NEEDLE GAITING

2. Interlock Gaiting: When needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at **face-to-face position** with each other is known as Interlock gaiting.

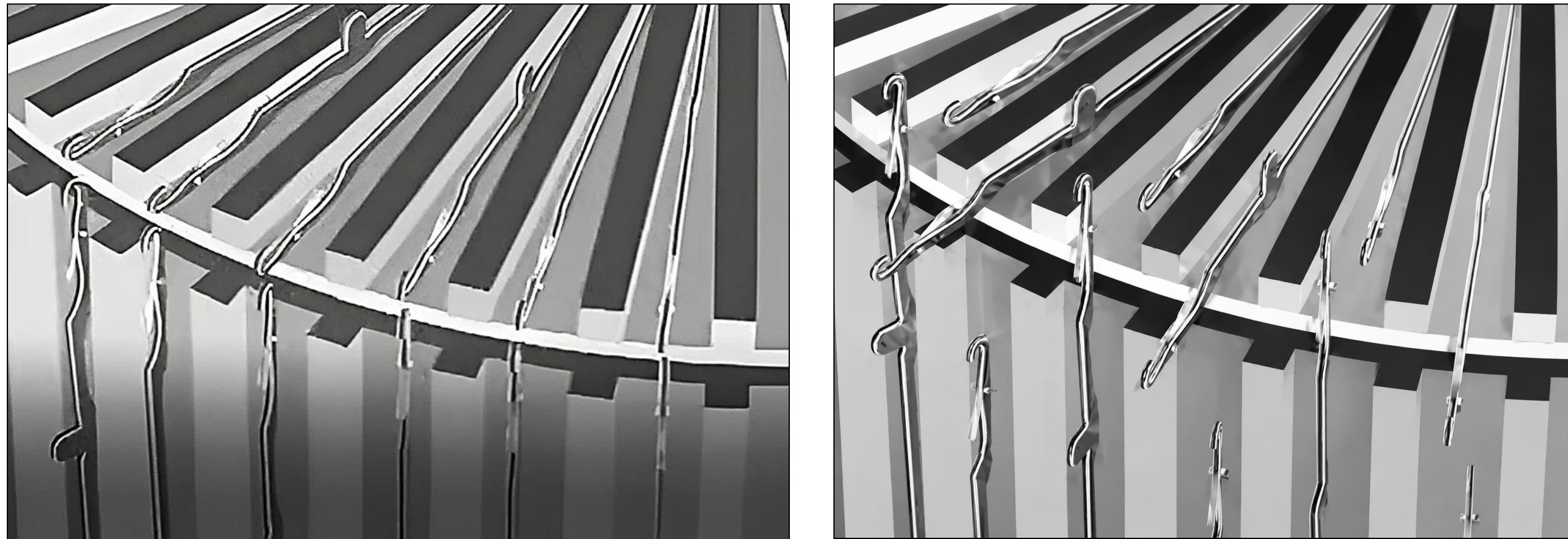
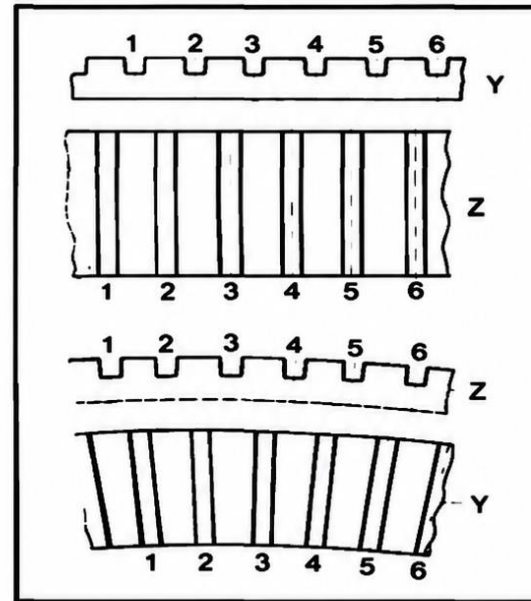
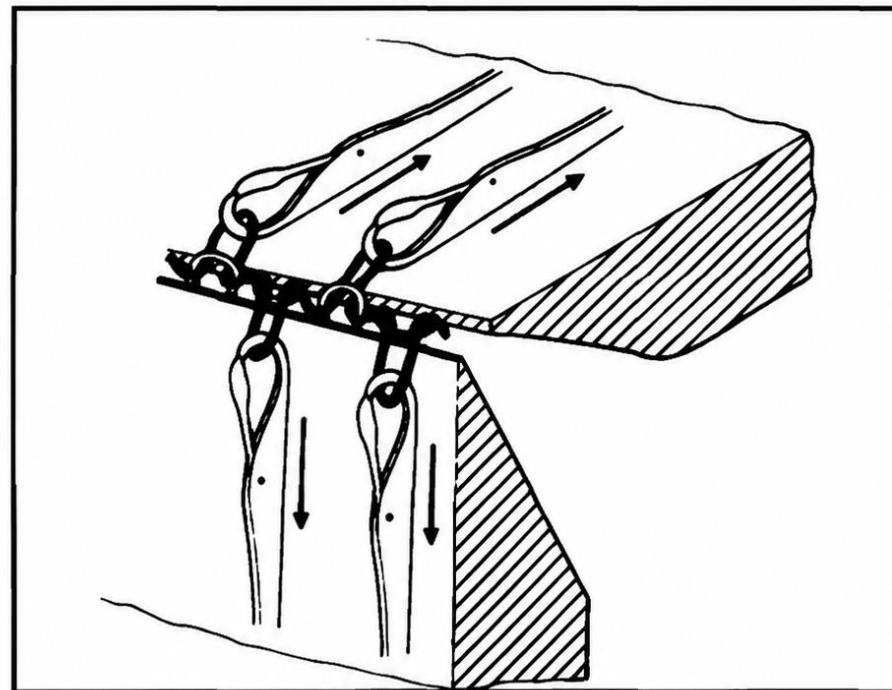


Fig: Interlock Gaiting

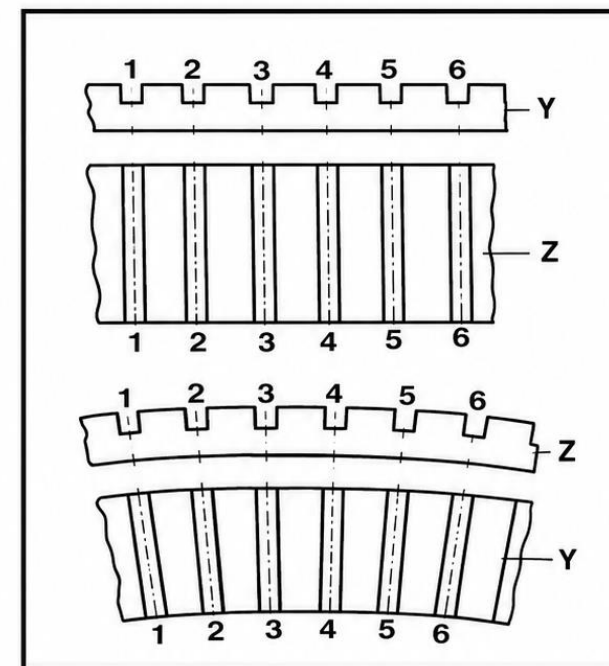
NEEDLE GAITING



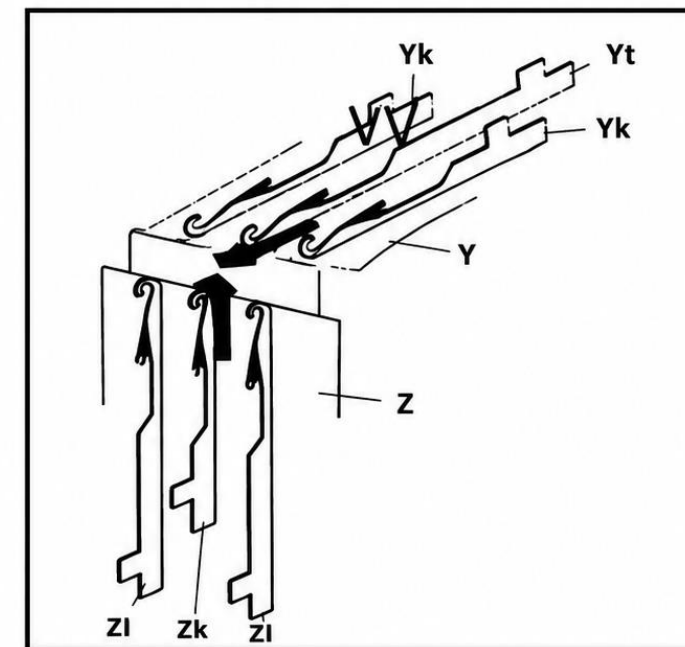
Rib gaiting for cylinder and dial on a rib circular knitting machine in front view (above) and top view (below)



Rib position of dial and cylinder needles



Interlock gaiting for cylinder and dial on a rib circular knitting machine in front view (above) and top view (below)



Interlock position of dial and cylinder needles

CONVERSION FROM RIB TO INTERLOCK GATING

- ❑ A rib knitting machine can be changed into an interlock machine if it has different types of needles in both the cylinder and dial, and the needles can be controlled separately.
- ❑ To convert it, the cylinder or dial is rotated by half a needle pitch so that the cylinder and dial needles come directly opposite each other.
- ❑ After this change, the machine works like a normal interlock knitting machine.
- ❑ In theory, an interlock machine can also be changed into a rib machine, *but this is not common because interlock machines usually have finer gauges.*
- ❑ If cylinder and dial needles cross too closely in rib settings, there may not be enough space, which can cause yarn damage or machine problems.
- ❑ Generally, interlock circular knitting machines use fine gauges (usually finer than E24).

NEEDLE TIMING

Needle timing is the *position difference between the dial needle knock-over point and the cylinder needle knock-over point* in a circular knitting machine.

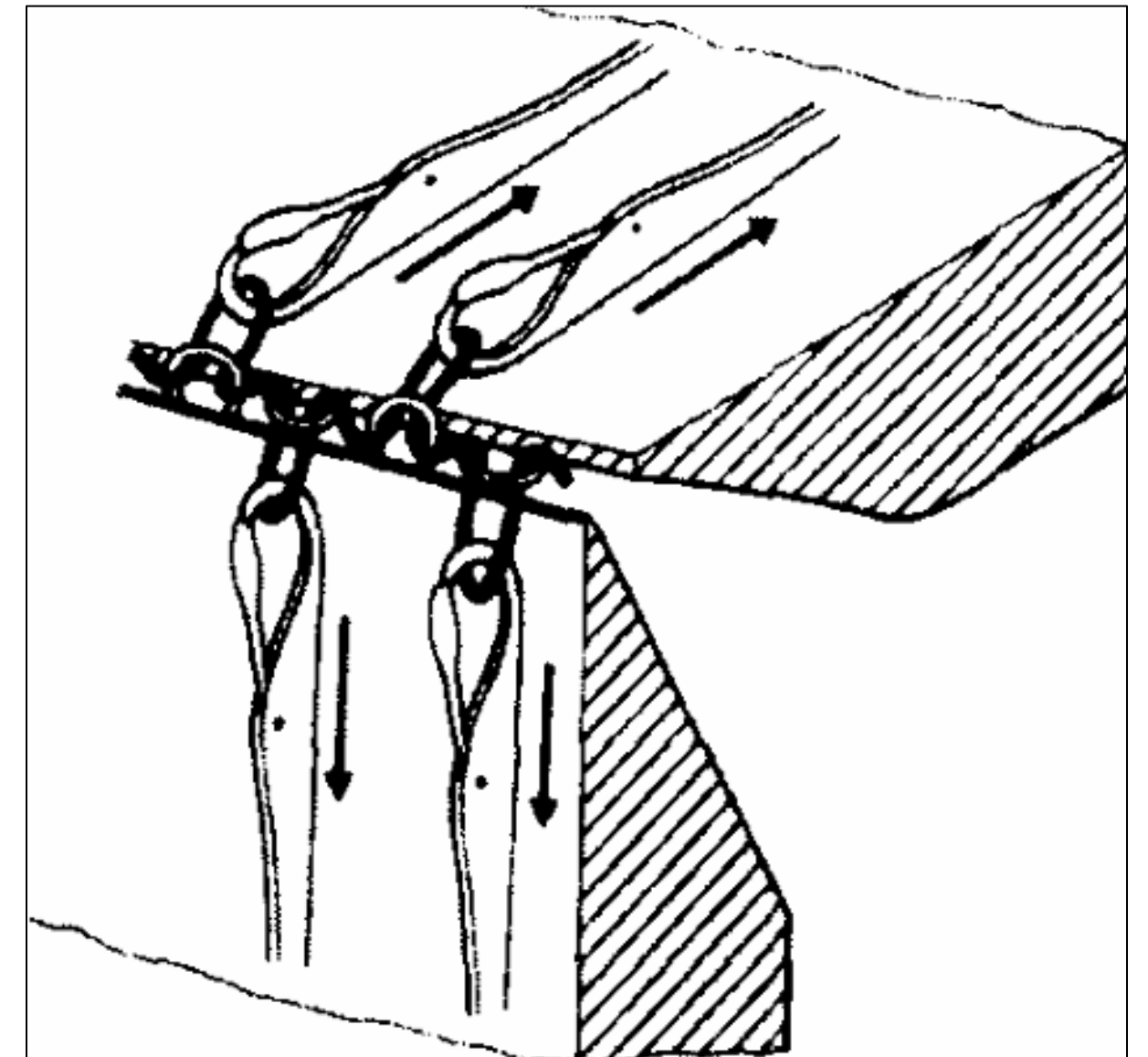
It is adjusted by changing the *position of the dial cam plate compared with the cylinder cam*. Individual adjustment can be made by changing or replacing the stitch cam profile.

There are two types of needle timing:

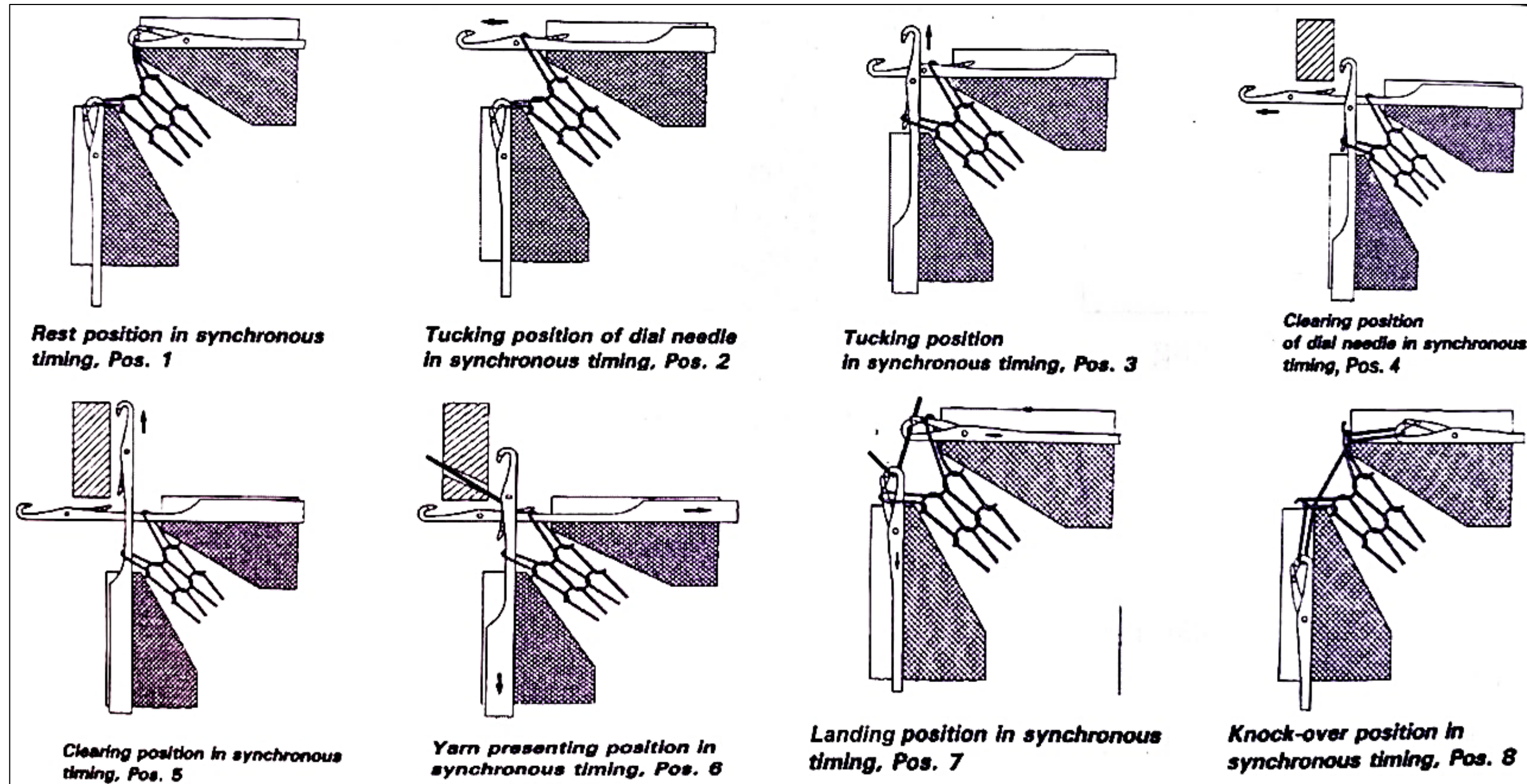
- ❑ Synchronized timing: *Dial and cylinder needles knock over at the same time.*
- ❑ Delayed timing: *Dial needle knock-over occurs later than the cylinder needle.*

SYNCHRONIZED TIMING

- ❖ In synchronized timing, the *cylinder needles and dial needles knock-over their loops at the same time.*
- ❖ Both needles reach the knock-over point together, and *their knock-over depths are equal.*
- ❖ During loop formation, the *yarn is pulled in two opposite directions by the cylinder and dial needles, which creates high tension in the yarn.*
- ❖ This timing can be used for *most rib and interlock structures.*
- ❖ Fabrics produced with synchronized timing are *usually looser and have uneven stitches* because of the high yarn tension.



KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN SYNCHRONOUS TIMING



KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN SYNCHRONOUS TIMING

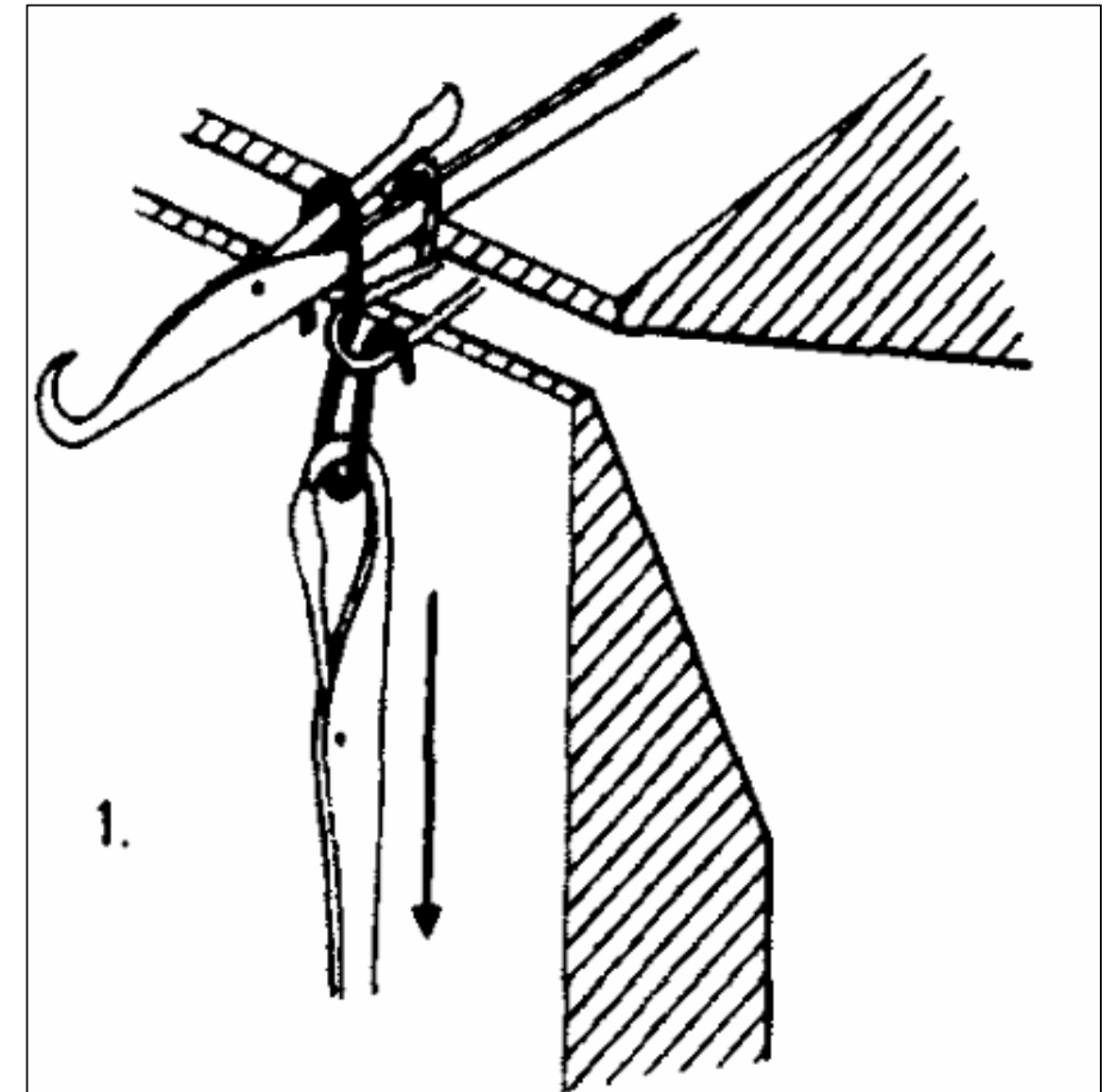
GROZ-BECKERT

Loop formation: double jersey

close-up with yarn guide

DELAYED TIMING

- ❖ In delayed timing, the *dial needles knock-over their loops after the cylinder needles*.
- ❖ The *cylinder needles first pull the loops* and *then rise slightly to reduce yarn tension*. After this, the *dial needles complete their knock-over process*.
- ❖ Because of this delay, the *dial loops become longer than the cylinder loops*.
- ❖ The fabric produced with delayed timing is:
 - ✓ Tighter in structure, even in stitch formation
 - ✓ More rigid and stable, heavier and wider
 - ✓ Produced with less strain on the yarn



DELAYED TIMING

Why are dial loops longer?

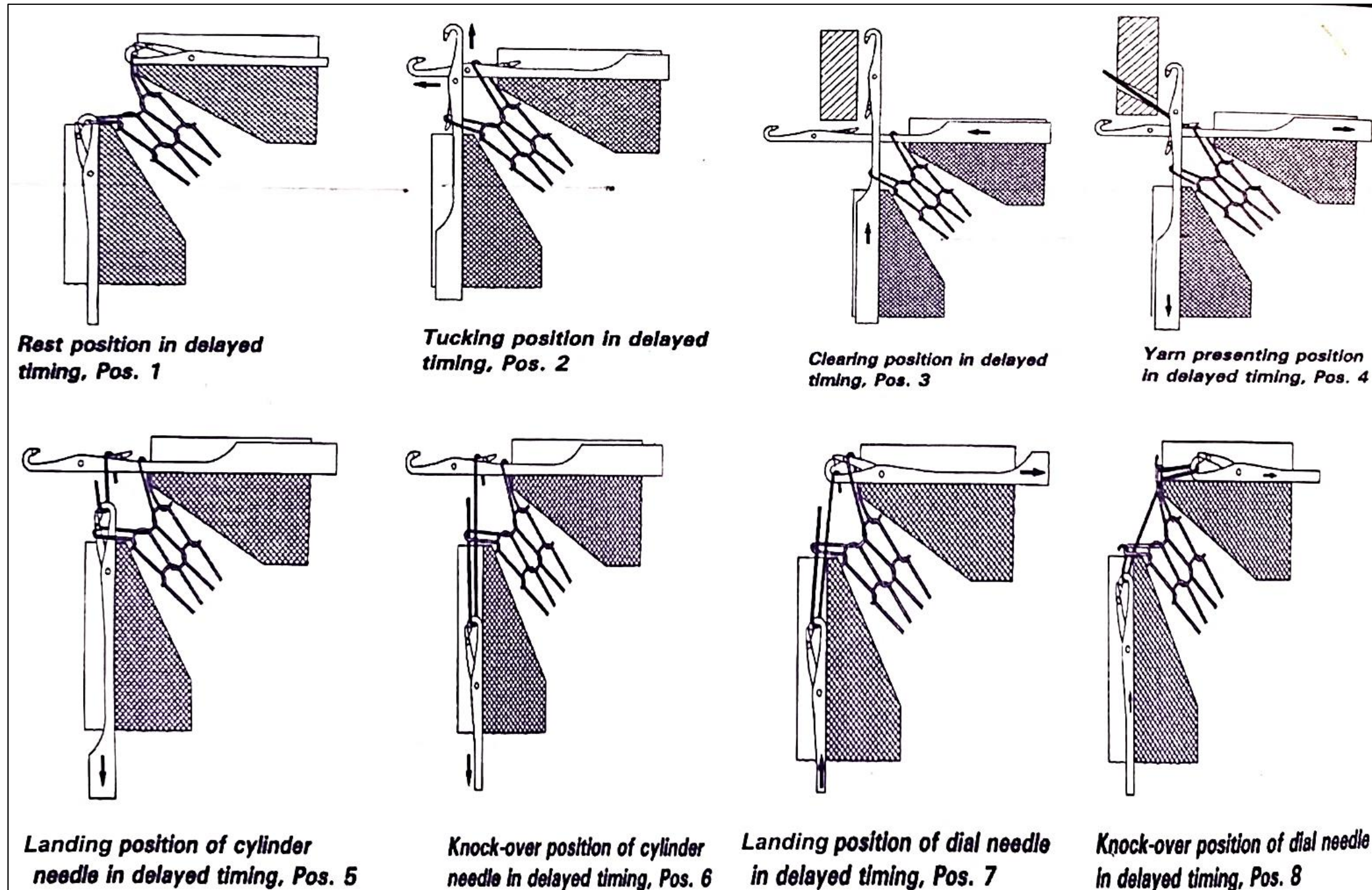
Because the dial needles have extra yarn available. During delayed timing: The cylinder needles have already drawn some yarn. The dial needles then pull their loops and receive a small amount of extra yarn ("robbed" from cylinder loops). Therefore, the dial loop length becomes larger.

Then how are stitches more-even?

Normally, in synchronized timing: Both needle sets pull the yarn at the same time. The yarn tension is very high and unequal forces can create irregular loops.

In delayed timing: Cylinder and dial needles work separately instead of pulling against each other at the same moment. The yarn tension is reduced. The loops form more smoothly and consistently. So, even stitches do not mean cylinder and dial loops are the same size. It means the stitches are more regular and stable.

KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN DELAYED TIMING



INTERLOCK CIRCULAR KNITTING MACHINE



INTERLOCK CIRCULAR KNITTING MACHINE

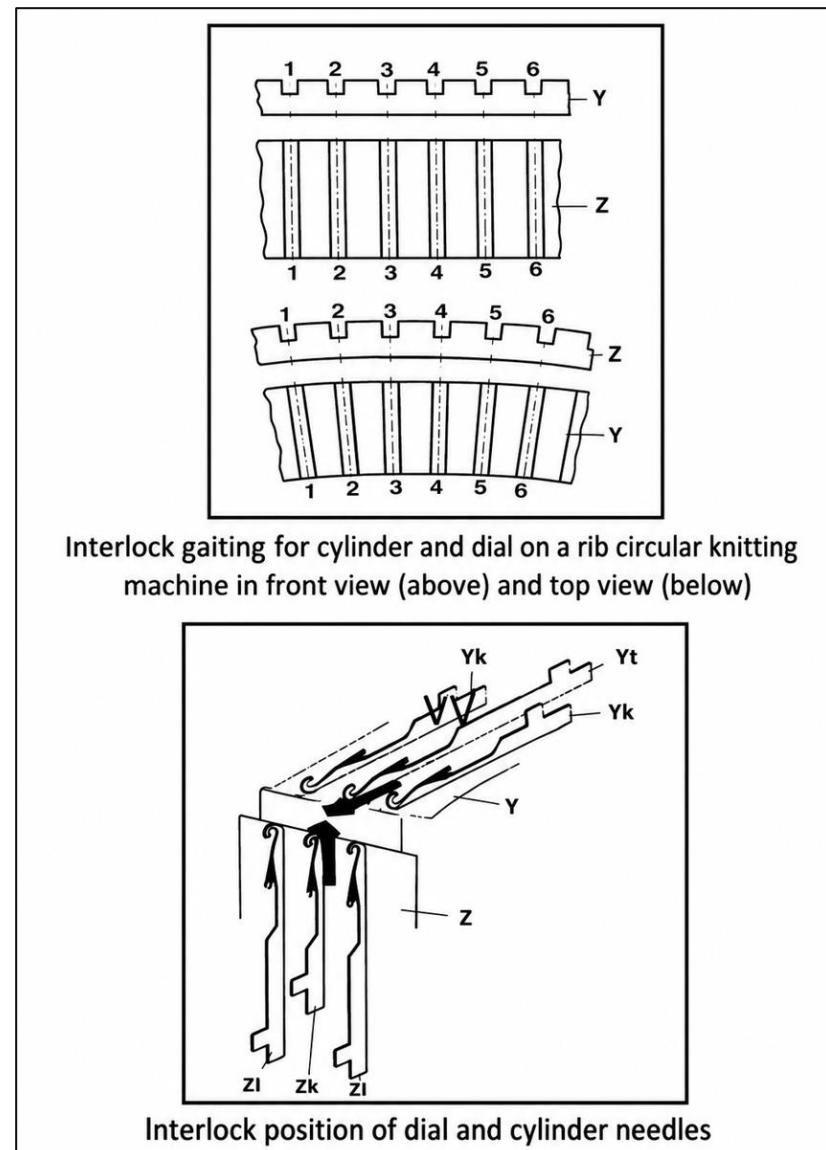
Interlock fabric is mainly produced on special cylinder and dial circular machines and some double-system Vee-bed flat machines.

Requirements of an Interlock Machine:

The *needles in the cylinder and dial must be exactly opposite each other*. This is called interlock gating. At each feeder, only **one set of needles works**, while **the other set follows a non-knitting path**. Each needle bed has two separate cam systems: One controls half of the needles at one feeder. The other controls the remaining needles at the next feeder. Needles are arranged in an alternate pattern, **so opposite needles do not knit together at the same feeder**. Conventional Interlock Machine uses two types of needles: Long needles & Short needles. **Long needles work at the first feeder. Short needles work at the second feeder. Needles are arranged alternately, with a long needle opposite a short needle.**

INTERLOCK CIRCULAR KNITTING MACHINE

At one feeder, long cylinder and dial needles form stitches. At the next feeder, short cylinder and dial needles form stitches. The needles not knitting at a feeder move through a non-knitting track. Modern machines usually use needles of the same length.



FEATURES OF INTERLOCK CIRCULAR KNITTING MACHINE

- ❖ Machine has two beds, machine may be flat or circular but generally circular.
- ❖ There are two subsets of needles in each bed, the subsets are known as short and long needles respectively.
- ❖ Two sets of cam system, cam path in each bed accommodate short and long needles in the other bed and vice-versa.
- ❖ Short needles in one bed make loops in conjunction with short needles in the other bed. Similarly, long needles in one bed make loops in conjunction with long needles in the other bed.
- ❖ Separate yarns are to be supplied to the short needles and long needles for loop formation through separate feeders.
- ❖ Short needles and long needles don't make loop simultaneously but with a time gap.
- ❖ Loops made by one set of needles are locked by the loops made by the other set of needles.
- ❖ Each interlock course is composed of two rib courses.
- ❖ Minimum two yarns are needed to produce a fabric.
- ❖ Fabric has a double face as well as a stable structure.
- ❖ Machine is complicated and costly.

CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Cylinder cam system:

The cylinder needle cam system consists of the followings:

- ❖ A, is a clearing cam which lifts the needle to clear the old loops.
- ❖ B and C, are the stitch and guard cams respectively and are vertically adjustable for varying stitch length.
- ❖ D, is upthrow cam, to raise cylinder needle whilst dial needle knocks over.
- ❖ E and F, are the guard cams, to complete the track.
- ❖ G and H, provide the track for the idling needles.

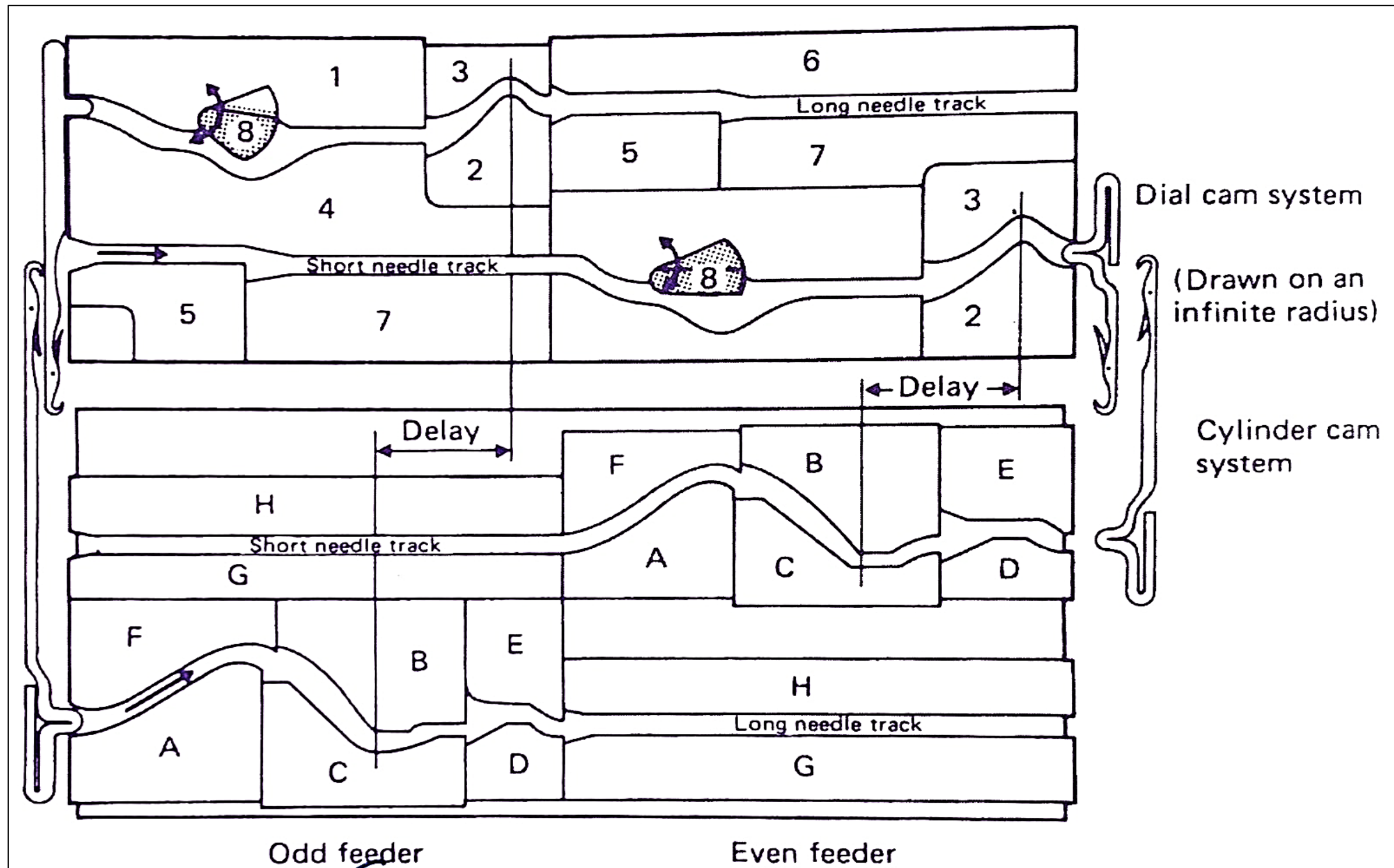
CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Dial cam system:

The dial needle cam system consists of the followings:

- ❖ 1, is a raising cam to the tuck position only.
- ❖ 2 and 3, are the adjustable dial knock-over cams.
- ❖ 4, is a guard cam to complete the track. (Drawn on an infinite radius) Cylinder cam system
- ❖ 5, is an auxiliary knock-over cam to prevent the dial needle re-entering old loop.
- ❖ 6 and 7, provide the track for the idling needles.
- ❖ 8, is a swing type clearing cam, which may occupy the knitting position as shown at feeder 1 or the tuck position as shown at feeder 2.

CAM SYSTEM OF INTERLOCK KNITTING MACHINE



PURL KNITTING MACHINE



PURL KNITTING MACHINE

In a flat links-links machine, double-headed latch needles are used. These needles can move between two needle beds because the slots of both beds are placed opposite each other.

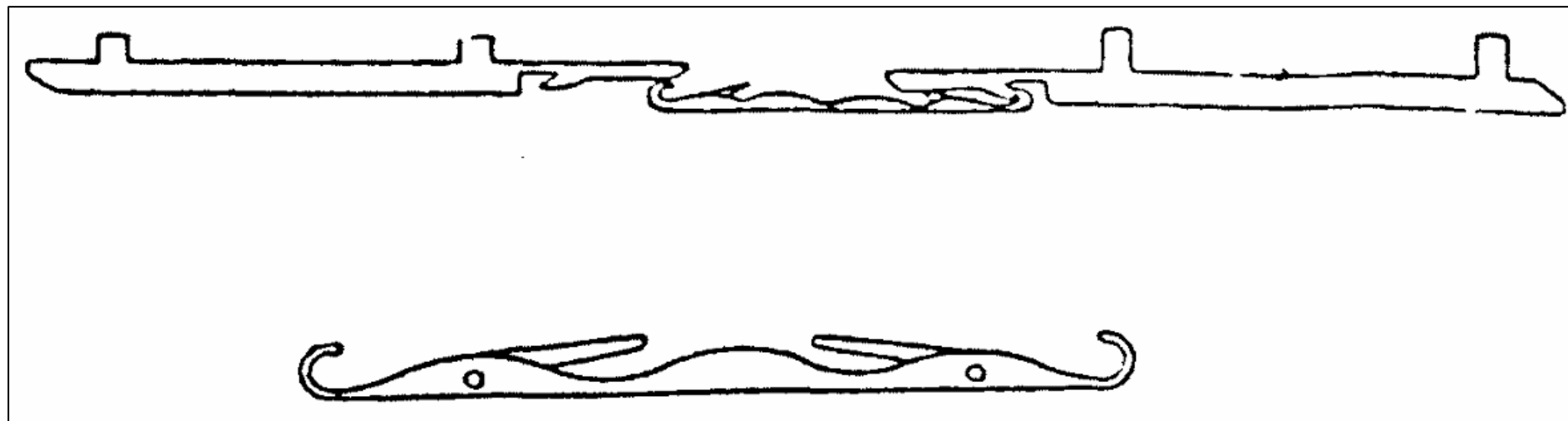
The needle beds have special knock-over teeth that hold the old loops during stitch formation.

Each needle has two sliders:

Slider M controls the left head of the needle.

Slider N controls the right head of the needle.

These sliders move the needle back and forth to form stitches.



FEATURES OF PURL KNITTING MACHINE

- ❖ Machine has two beds. Machine may be flat or circular.
- ❖ There is only one set of needles which change the bed in alternatively knitting cycle.
- ❖ Needles are of a special type; generally double hooked latch needles are used.
- ❖ Needles are shifted from one bed to another bed with the help of sliders. Two sliders are needed for each needle.
- ❖ Same needle makes face loop in one bed a back loop in other bed.
- ❖ Alternative courses are made of all face loops and all back loops. As a result, each wale is made of face loop and back loop in an alternative order.
- ❖ Fabric is reversible in appearance and has a soft handle.
- ❖ Fabric has very high extensibility in the length direction, making it suitable for kidswear.
- ❖ Fabric does not curl at the edges because of the alternate face and back loop courses.
- ❖ Machines are very rare.

PURL KNITTING MACHINE

Stitch Formation Process

1. Needle Movement and Loop Transfer

- ☐ A slider moves the needle from one bed to the other.
- ☐ The old loop opens the needle latch and moves to the middle of the needle.

2. Yarn Laying

- ☐ The yarn is placed on the needle hook.
- ☐ The yarn must pass under the hook correctly and should not be caught by the latch.

3. Underlapping and Pressing

- ☐ The needle moves forward.
- ☐ The old loop is held by the knock-over teeth.
- ☐ The old loop closes the latch, trapping the new yarn inside the needle hook.

PURL KNITTING MACHINE

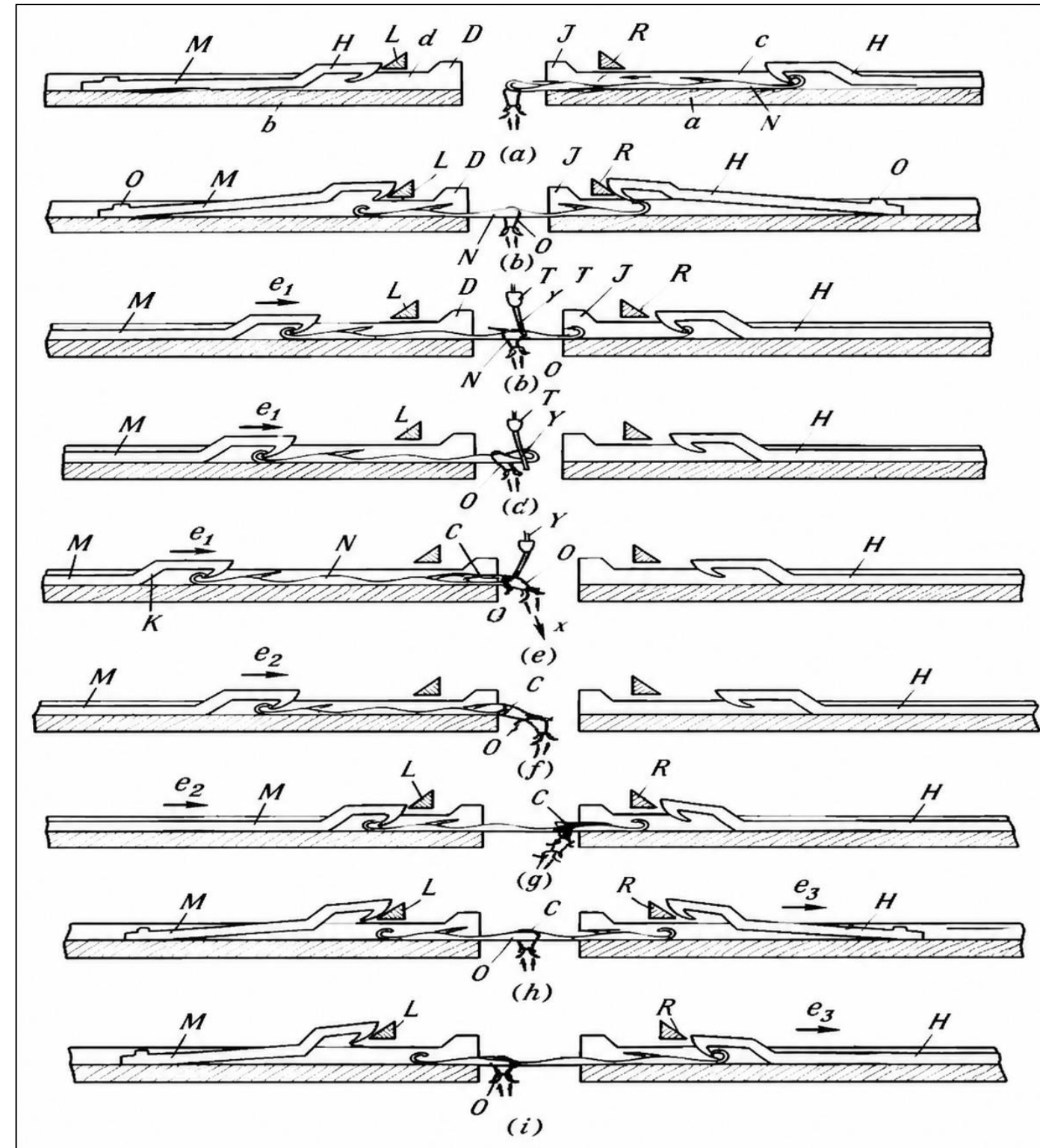
4. Landing, Joining, and Casting-off

- ☐ The needle continues moving.
- ☐ The old loop moves over the latch and joins with the new yarn.
- ☐ The old loop is removed, and a new loop is formed.

5. Formation of Purl Stitch

- ☐ When the left needle head forms the loop, the old loop is cast off to the left side.
- ☐ When the right needle head forms the loop, the old loop is cast off to the right side.
- ☐ This opposite loop formation creates a purl stitch.

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GROZ-BECKERT

Loop formation: links and links technology

Needle and slider movement

THANK
YOU !

